

Safe Assets in Emerging Market Economies

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Abstract

Do local-currency sovereign bonds in emerging markets (EMs) work as safe havens? Using data from fourteen high-income EMEs, we estimate the convenience yields of these bonds in domestic currency and dollar terms. The domestic safety premium in these economies is larger in magnitude than the liquidity premium. We test their safe-haven status by estimating the convenience yields' response to global and country-specific risk factors. The empirical analysis reveals that local-currency sovereign bonds serve as safe havens against country-specific risks in both domestic currency and dollar terms; against global risks, the fall in convenience yields is explained by the liquidity premium.

JEL Codes: E43, F30, G12.

Keywords: convenience yields, safe haven, sovereign bonds, emerging markets.

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1 Introduction

What asset works as a safe haven in an emerging economy? Is it a foreign asset or a local-currency sovereign bond? Or both, but conditional on the type of shock? In this paper, we answer these questions by estimating convenience yields on EME sovereign bonds and employing their response to global and country-specific uncertainty to measure their use as a safe haven. “Convenience yields” capture the value for investors of the non-pecuniary benefits of an asset’s liquidity and safety beyond the value of discounted future cash flows (Krishnamurthy and Vissing-Jorgensen, 2012). We show that if an asset works as a safe haven, its safety or liquidity value should increase with systematic risk. Over the past two decades, EME governments have deepened their local currency bond markets and improved their credit ratings. Local currency debt now represents the lion’s share of outstanding sovereign bonds in EMEs (BIS, 2020). This paper is the first attempt to expand our knowledge of safe assets by studying the convenience yields of local-currency sovereign bonds across fourteen high-income EMEs¹, understand their interactions with shocks of different natures, and contrast them with those of advanced economies.

There is substantial evidence of convenience yields on sovereign bonds in advanced economies and their role as safe assets. The safe asset status of government bonds influences equilibrium interest rates, expands government fiscal capacity, and acts as a transmission channel for large-scale central bank asset purchases². This paper underscores that, in emerging markets, government bonds can function as local safe assets, albeit competing with other global sources of safety. Recent work by Kekre and Lenel (2024) suggests that shifts in safety and liquidity demand contribute significantly to output volatility in the U.S. and globally.

How to think about safety in the context of EMEs? This paper will refer to *relative* safety rather than absolute safety. These sovereign bonds will carry risks, including credit, liquidity, and currency risks. However, they will carry a convenience yield if they are *safer* than the relevant alternatives (which will be described throughout the paper), and if investors have a

¹Brazil, Chile, Colombia, Hungary, Indonesia, Korea, Malaysia, México, Perú, Philippines, Poland, South Africa, Thailand, and Turkey.

²See Del Negro et al. (2017a), Lenel, Piazzesi, and Schneider (2019), Jiang et al. (2022), Krishnamurthy and Vissing-Jorgensen (2011).

special demand for this lower risk³. For example, a financial intermediary with an inelastic demand for assets that offer higher nominal repayment or an asset that can be easily sold.

The convenience yields we will estimate may stem from safety or liquidity services, two distinct yet intertwined concepts. Liquidity pertains to the ease of selling assets for cash, while safety refers to an asset’s valuation at face value without extensive analysis (Gorton, 2017). A country-by-country analysis of local financial markets allows us to capture the safety premium by comparing assets with same liquidity but different credit risk, and the liquidity premium by comparing assets with the same credit risk but different liquidity⁴.

In the first part of the paper, we provide two estimations of convenience yields of EME local-currency sovereign bonds. The first, which we call the “domestic convenience yield”, compares the local-currency sovereign bond to a domestic private local-currency asset with similar maturity but lacking equivalent safety and liquidity services. We provide evidence that in most countries in the sample, government bonds are at the center of the local financial market functioning and liquidity. As shown in Krishnamurthy and Vissing-Jorgensen (2012), the spread between these assets, after controlling for risk premia, gauges the safety and liquidity premium on the local-currency sovereign bond. This measure enables the disentanglement of the convenience yield between a safety and a liquidity premium. We obtain the safety premium by comparing unsecured interbank loans with collateralized interest rate swaps. The liquidity premium is obtained by comparing the latter with the yield of a government bond. Data is available for assets with maturities of 3 months and 1 year.

The second estimate, which we call the “dollar convenience yield”, compares a synthetic dollar bond (a local-currency EME sovereign bond with its cash flows swapped into dollars via a forward contract) against non-Treasury-safe dollar bonds (such as highly rated U.S. corporate or U.S. agency bonds). This is the convenience yield of the local-currency sovereign bond in dollar terms, not the bond denominated in foreign currency. This can be a relevant

³“Risk” will be used throughout the paper in two dimensions. First, in Section 2, when we estimate the convenience yields, we will control for the credit, liquidity, currency, and regulatory *risk* of the sovereign bond (these are idiosyncratic to the asset and can confound the estimation of the convenience yield). In Sections 3 and 4, we will test the response of convenience yields to systematic sources of *risk* (recessions, financial distress, etc.). These are country-wide or global sources of risk that can affect sovereign bonds (and their idiosyncratic components).

⁴Chaumont (2021) and Passadore and Xu (2022) provide models that characterize how credit risk and liquidity interact in secondary markets.

measure for an investor who decides on a portfolio of domestic and foreign currency assets and measures her returns in dollars⁵. EME sovereign bonds will have either higher credit ratings or higher liquidity than some of these assets, justifying the existence of a convenience yield. Since both bonds are in dollars, the spread does not include currency risk, which allows us to build on the methodology used by Du and Schreger (2016) and Du, Im, and Schreger (2018). We show that the spread between the two assets, in this case, is the sum of (1) the differential default risk, (2) regulatory risk, or the risk of losses produced by regulations imposed by the EME government (such as taxes on capital outflows or currency convertibility restrictions), (3) the covariance of the local currency with these risks, (4) frictions in bond and forward currency markets, and (5) the differential convenience yield. After accounting for the first four, we obtain the latter as a residual. Data is available for assets with a 5-year maturity.

The analysis reveals a significant convenience yield for EME local-currency sovereign bonds, which is robust for both the domestic and dollar convenience yields. Consistent with convenience yields arising from safety or liquidity services, results show that convenience yields are increasing in the level of the monetary policy rate, reflecting the “money-like” properties of these sovereign bonds, as in Nagel (2016); and decrease in the supply of government debt, reflecting a downward-sloping demand curve for safety, as in Krishnamurthy and Vissing-Jorgensen (2012). Regarding the domestic convenience yield, the safety premium is much larger in magnitude than the liquidity premium. The former has its highest value during the 2008 crisis but sharply declines during the Covid pandemic. The latter is positive and spikes in the run-up to the 2008 crisis, but then is close to zero until 2015 when monetary policy started to tighten. The dollar measure indicates an average of nearly 30 basis points, and we argue it represents mostly a liquidity premium.

To assess the sovereign bonds’ role as safe havens, we propose a framework that links the response of an asset’s convenience yield to different sources of systematic risk to its safe haven status. In particular, an asset serves as a safe haven against global risk if its safety or liquidity premium increases in response to the global risk factor. Similarly, an asset

⁵For example, foreign investors have increased their participation in EME local-currency sovereign bonds in recent years. See Onen, Shin, and von Peter (2023) and Du and Schreger (2022).

serves as a safe haven against local or country-specific risk if its safety or liquidity premium increases with the local risk factor. The global factor captures investors’ risk sentiment on the global economy (Rey, 2013; Miranda-Agrippino and Rey, 2022). The local factor analogously captures investors’ risk sentiment towards a given country or country-specific frictions that can arise from economic policy uncertainty affecting investors⁶.

We test this prediction empirically by estimating the response of convenience yields to different sources of systematic risk. We find that both the dollar and the domestic convenience yield (the safety and the liquidity premium) increase against country-specific risk, as captured by the Economic Policy Uncertainty (EPU) index of Baker, Bloom, and Davies (2016) or the economic and financial risk index of the International Country Risk Guide (ICRG)⁷. However, the results are markedly different for global risk⁸. Higher global risk sentiment, as captured by the VIX or the global factor in risky asset prices of Miranda-Agrippino and Rey (2020), is associated with a lower dollar convenience yield, a lower domestic liquidity premium, but with a *higher* domestic safety premium, suggesting that the sovereign bond can be a safe alternative compared to other domestic private alternatives.

To gain further insight, we analyze two exogenous shocks to EMEs: the Taper Tantrum (which signaled the end of dollar liquidity supply via the end of the Fed’s large-scale asset purchases) and the Covid pandemic (which triggered a global flight to safety). While the dollar convenience yield increased amid scarcer liquidity during the Taper Tantrum, it significantly dropped against the flight to safety in March 2020. Surprisingly, this drop was not mechanically driven by the rise in credit risk or higher risk aversion but by a switch in investors’ preferences away from EME bonds and towards global safe assets.

The results have several implications for EMEs. If sovereign bonds are valued for their safety and liquidity services, then the commonly held view that yield-oriented investors

⁶This paper does not consider the choice between safe havens or multiple equilibria. Rather, it characterizes how, given that a bond is considered a safe haven, the convenience yield will correlate positively with those systematic risks against which it is perceived as safe. The endogenous evolution of a safe haven status is analyzed in, for example, He, Krishnamurthy, and Milbradt (2019) and Brunnermeier, Merkel, and Sannikov (2024).

⁷Kalemli-Ozcan and Varela (2023) show that these variables capture a relevant factor explaining systematic risk in emerging market economies.

⁸González-Rozada and Levy Yeyati (2008) find that a large fraction of the time variability of EME bond spreads is explained by the evolution of global factors such as risk appetite, and global liquidity.

determine capital inflows to EMEs might not always hold. In addition, the responses of the convenience yield against global and local risk factors suggest that their safe asset status depends on the nature of the shock and the benchmark asset. They are a safe haven compared to domestic private assets against local and global systematic risks. However, they are a safe haven compared to foreign private assets only against local risk. Suppose we take the domestic and the dollar convenience yield as coming from a domestic and a global investor, respectively. In that case, EMEs should pay attention to their investor composition as it can add an extra source of volatility and fiscal strain during episodes of global uncertainty.

Related Literature. The empirical literature is ample in the study of the safety of U.S. Treasuries against comparable dollar private debt (Krishnamurthy and Vissing-Jorgensen, 2012; Greenwood, Hanson, and Stein, 2015; Nagel, 2016) and against sovereign bonds of other advanced countries (Du, Im, and Schreger, 2018; Jiang et al., 2021). Diamond and Van Tassel (2023) estimate convenience yields using domestic assets for G10 countries, which have some similarities with the domestic convenience yield we estimate. This paper is the first attempt to apply this empirical work to pricing local-currency sovereign bonds in emerging markets. This exercise can contribute to the literature in two ways. First, moving beyond advanced economies can expand the cross-section of convenience yields and help us better understand their determinants and dynamics. Second, moving beyond the study of global safe assets can help us discover the role of “local” safe assets, which serve a purpose in mitigating country-specific shocks.

The literature has offered many rationales for the pricing of a convenience yield: insurance provision against idiosyncratic shocks (Brunnermeier, Merkel, and Sannikov, 2024), liquidity regulations and financial repression (Chari, Dovis, and Kehoe, 2020; Payne and Szoke, 2024), bonds’ use as collateral (Devereux, Engel, and Wu, 2023; Mendoza and Quadrini, 2023). In this paper, we study convenience yields and their dynamics against heightened systematic risk, but they are, in principle, compatible with any of these explanations.

Although the methodologies and data differ, the empirical results of this paper can be reconciled with recent work estimating the demand for sovereign debt in EMEs (Fang, Hardy, and Lewis, 2022; Zhou, 2024). As discussed later, these papers find a more stable and inelastic demand for certain foreign holders of EME sovereign debt, which could be consistent

with a convenience yield. Similarly, the model complements recent attempts to introduce a liquidity role for government bonds in banks’ balance sheets in EMEs (Perez, 2018) and the external finance cost of global intermediaries who lend to EME sovereign bonds (Morelli, Ottonello, and Perez, 2022). Although they do not explicitly introduce convenience yields, they could rationalize the existence of a liquidity premium for domestic banks and a “bond inconvenience” for global banks in response to global shocks, respectively.

Convenience yields are important in explaining exchange rate levels and puzzles (Engel, 2016; Jiang, Krishnamurthy, and Lustig, 2021; Engel and Wu, 2023); can lower equilibrium interest rates (Del Negro et al., 2017a); can increase the government’s fiscal capacity (Jiang et al., 2022); and are a channel for the effectiveness of large scale asset purchases (Krishnamurthy and Vissing-Jorgensen, 2011; Del Negro et al, 2017b). Moreover, they are a significant driver of global capital flows, as noted in Jiang, Krishnamurthy, and Lustig (2019) and Kekre and Lenel (2024). The results of this paper can provide valuable input for studying these policies in EMEs.

This paper also adds to recent studies of the drivers of capital inflows to EMEs. While the conventional view has mostly assumed they are driven by yield-oriented investors that respond to interest rate differentials with the U.S., recent papers have found that risk perceptions also play a significant role. Moreover, US policy seems to affect these risk perceptions in emerging markets (Kalemli-Ozcan, 2019; De Leo, Gopinath, and Kalemli-Ozcan, 2022; Kalemli-Ozcan and Varela, 2023). This paper complements these recent findings, as the convenience yields we estimate can be considered part of the risk perceptions and explain part of the uncovered interest parity deviations studied in those papers.

Finally, by documenting the existence of domestic safe assets and their interaction with the U.S. Treasury, this paper contributes to the literature on safe assets shortages (Caballero, Farhi, and Gourinchas, 2016, 2017). Recently, Mendoza and Quadrini (2023) quantified how the reliance on U.S. debt as the sole source of safety has increased global financial instability. The line of work of our paper could contribute to the questions of what is required to expand the supply of safe assets or reduce the global demand for U.S. safe assets.

The paper proceeds as follows. Section 2 presents the methodology for estimating convenience yields. Section 3 analyzes the results and provides further evidence that the estimated

convenience yields come from safety and liquidity services. Section 4 tests the bonds' role as safe havens. Section 5 concludes.

2 Methodology

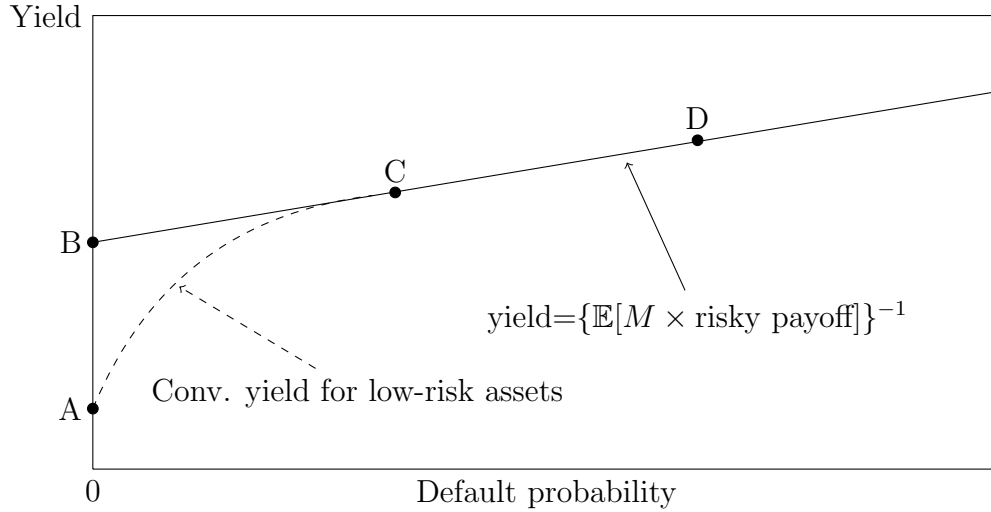
In this Section, we outline a framework that extends the methodology in Krishnamurthy and Vissing-Jorgensen (2012) in the direction of including assets with some level of idiosyncratic risk and assets in different currencies. Similar to that paper, an asset earns a convenience yield if (1) it has lower credit risk or higher liquidity than an alternative asset with the same cash flows and maturity, and (2) investors derive non-pecuniary benefits from that lower risk (because of a preferred-habitat motive, collateral demand, etc.).

The countries included in the sample are Brazil, Chile, Colombia, Hungary, Indonesia, Korea, Malaysia, Mexico, Peru, Poland, the Philippines, Thailand, Turkey, and South Africa. The Appendix shows that, in each of these countries, the large majority of all outstanding sovereign bonds are denominated in their domestic currency.

We begin by providing intuition on why convenience yields differ from standard risk premia. The latter has been widely studied in the context of emerging markets, while the former has not. Figure 1 shows the distinction between convenience yields and the risk premium in a standard asset pricing model, considering the case of credit risk (an analogous explanation could be given using liquidity risk). The straight line represents the yield of a risky bond as determined in a consumption-based capital asset pricing model. As default risk rises (to the right on the horizontal axis), the yield on the bond rises. It is common in the literature to speak of a “safety premium” when comparing, say, two bonds, C and D , to refer to their spread.

The convenience yield refers to something different. For bonds that have very low default risk, the yield decreases as a function of the bond's safety, more so than predicted by the standard asset pricing kernel (the dashed line in the Figure). Why? There might be investors with an inelastic demand for assets that promise a stable nominal repayment, or that can be easily sold, and they derive non-pecuniary benefits from holding them. In the Figure, point B represents the risk-free rate, and point A is the yield on a bond that provides non-

Figure 1: Yields, Risk Premia, and Convenience Yields



Notes: Figure illustrates the difference between risk premia (spread between bonds *C* and *D*, given by a standard C-CAPM model between two bonds with different risk -the straight line), and convenience yields (spread between bonds *B* -the risk free asset- and *A*, due to non-pecuniary services of *A* that lower its yield below the C-CAPM model -the dashed line).

pecuniary services due to its low risk. The vertical distance between the two illustrates the convenience yield, which can be captured within the standard model in a reduced-form way by including bond holdings in the investor's utility function. Since, in this case, *B* and *A* have the same credit risk, this convenience yield might come from the larger liquidity of asset *A*. Analogously, the convenience yield of two assets with equal liquidity represents a safety premium.

We provide two measures of the convenience yield, which differ in the “alternative asset” used in the comparison (point *B* in Figure 1). In the first, we compare the local-currency sovereign bond against a private domestic asset with higher credit risk and/or lower liquidity. The second comparison is between the local-currency sovereign bond and a private foreign dollar asset with higher credit risk and/or lower liquidity.

The first could be considered capturing an investor's valuation - call her *d* - who compares the returns of only domestic assets, both financed with the same currency. The second could be considered capturing an investor's valuation -call her *f*- who measures returns in dollars and compares the local-currency sovereign bond with a non-Treasury safe dollar bond (an agency or a highly-rated corporate bond)⁹.

⁹Notice that the residence of the investor could go either way. The domestic convenience yield could come

For any of these investors, the framework modifies a standard representative agent model to include a term that captures agents deriving utility directly from holding a “convenience” asset. A representative investor maximizes

$$E \sum_{t=1}^{\infty} \beta^t u(c_t + \nu(\theta_t^i, \text{GDP}_t^i)) \quad (1)$$

where $i \in \{d, f\}$ denotes the investor. c_t is consumption from an endowment stream, and the second term represents the “convenience” benefits of holding bonds that provide safety or liquidity services, θ_t^i . The assets that enter into θ_t^i will be specified later for each investor. The agent’s income is GDP_t^i , measured in real terms. Problem (1) shows that both investors differ in the assets held in their portfolio and might differ in the endowment stream they receive, GDP_t^i .

The function $\nu(\cdot)$ is a reduced-form way of capturing non-pecuniary benefits from the safety and liquidity of certain bonds. For example, there are benefits of holding a liquid asset that eases transactions (as collateral) or having an asset that promises stable nominal repayments. These assets could be money, sovereign bonds, or private assets that share, to some extent, these characteristics (like insured bank deposits, central bank reserves, or corporate bonds of highly rated companies).

Assume that the convenience function is homogeneous of degree one in GDP_t^i and θ_t^i . Thus define $v(\frac{\theta_t^i}{\text{GDP}_t^i})\text{GDP}_t^i \equiv \nu(\theta_t^i, \text{GDP}_t^i)$. Assume that the convenience function is increasing in $\theta_t^i/\text{GDP}_t^i$, but the marginal convenience benefit is decreasing in $\theta_t^i/\text{GDP}_t^i$, and $\lim_{\theta_t^i/\text{GDP}_t^i \rightarrow \infty} v'(\theta_t^i/\text{GDP}_t^i) = 0$.

The Euler equation for holdings of a convenience asset, θ_t^i , gives the following expression for its price, P_t (to simplify, assume no default risk, which will be introduced later):

$$P_t = E_t[M_{t+1}P_{t+1}\Lambda_{t+1}^i]$$

where $M_{t+1} = \beta \frac{u'(C_{t+1})}{u'(C_t)} \frac{Q_t}{Q_{t+1}}$ is the pricing kernel and Q_t is the price level. $\Lambda_t^i \equiv 1/(1 -$

from a domestic investor who measures her returns in the local currency or from a foreign investor with a portfolio of local assets. The dollar convenience yield could originate from a foreign investor with investment opportunities in EMEs or from a domestic investor with investment opportunities in the US. The possibility of market segmentation -which breaks this equivalence- will be discussed in subsection 2.1.2.

$v'(\theta_t^i/\text{GDP}_t^i)$) captures the marginal benefits investor i derives from these bonds. A positive marginal value of convenience by investor i , $v'(\cdot)$, raises Λ_t^i , and therefore raises the price of the bond, P_t .

2.1 Domestic convenience yield

Consider investor d with the alternative of investing in a domestic sovereign bond (with some level of default risk) and a domestic private asset, but with higher risk and thus lower convenience services, such as a term deposit at a local commercial bank.

The portfolio of convenience assets of this investor is given by:

$$\theta_t^d = \theta_t^M + \kappa_t^{T,d} \theta_t^T + \kappa_t^{P,d} \theta_t^P \quad (2)$$

where θ_t^M , θ_t^T , and θ_t^P correspond to holdings of money or cash (as the most liquid domestic asset), sovereign bonds, and alternative private substitutes, respectively. The latter two are of the same maturity. $\kappa_t^{T,d}$ and $\kappa_t^{P,d}$ represent the investor's d relative preference for the convenience service of assets other than money. Both are assumed to be less than one. Time variation in $\kappa_t^{T,d}$ and $\kappa_t^{P,d}$ could come from changes in their safe asset status: if, in certain states of the world, investors switch preferences to the safety and liquidity services of other assets, beyond what would be explained by variations in credit risk.

Proposition 1. *The spread between the yield of a local-currency domestic private asset, y_t^P , and the yield of a local-currency sovereign bond, y_t^T , of the same maturity, can be decomposed as follows:*

$$y_t^P - y_t^T \approx (\lambda_t^{T,d} - \lambda_t^{P,d}) + (l_t^P - l_t^T) + (\xi_t^{T,d} - \xi_t^{P,d}) \quad (3)$$

where $\lambda_t^{T,d}$ measures the marginal safety/liquidity services the investor d derives from the sovereign bond (the convenience yield), $\lambda_t^{T,d} \approx \kappa_t^{T,d} v'(\theta_t^d/\text{GDP}_t^d)$, and $\lambda_t^{P,d} \approx \kappa_t^{P,d} v'(\theta_t^d/\text{GDP}_t^d)$; l_t^P and l_t^T are, for each asset, the expected default plus a risk premium associated with the covariance between default and the stochastic discount factor; and $\xi_t^{T,d}$, $\xi_t^{P,d}$ are the covariances between credit risk and the convenience yield.

Proof: see the Internet Appendix IA.

The domestic measure of the convenience yield of local-currency sovereign bonds is given by:

$$CY_t^d = \lambda_t^{T,d} - \lambda_t^{P,d} = (\kappa_t^{T,d} - \kappa_t^{P,d})v'(\theta_t^d/\text{GDP}_t^d) \quad (4)$$

Equation 4 shows that what we call the “domestic convenience yield” is really a *differential* convenience yield because we allow the domestic private asset to provide some (but lower) convenience benefits. Henceforth, whenever we mention the “domestic convenience yield”, we will refer to this differential convenience yield.

What increases the convenience yield on sovereign bonds, CY_t^d ? First, a lower supply of government debt, $\theta_t^T/\text{GDP}_t^d$, or a lower supply of substitutes, $\theta_t^P/\text{GDP}_t^d$. Second, if sovereign bonds provide the same liquidity services as money, then a lower supply of $\theta_t^M/\text{GDP}_t^d$ will also increase the convenience yield. The reduction (increase) in the supply of any of these assets would move the dashed line in Figure 1 down (up). Third, variations in the relative convenience service. For example, losing the safe asset status (switch in investors’ preferences towards other safe assets) can reduce $\kappa_t^{T,d}$. Regarding Figure 1, an increase (decrease) in $\kappa_t^{T,d}$ would move the dashed line for the sovereign bond further below (up) and away (closer) from the straight line.

How will we recover CY_t^d in the data? We will take data on spreads between the two types of assets, $y_t^P - y_t^T$, as a measure of the convenience yield of local-currency sovereign bonds. Following the insight from Krishnamurthy and Vissing-Jorgensen (2012), we compare assets with similar liquidity but different credit risk to capture the safety premia, and two assets with similar credit risk but different liquidity to capture the liquidity premia.

Appendix B provides a country-by-country description of the liquidity and credit risk associated with different instruments in the domestic currency of the local financial market, which we summarize here. For the liquidity premium, we use the spread between domestic currency interest rate swaps and the yield on local currency government bonds of the same maturity. In all the countries considered, government bonds play a central role as a source of market liquidity. Most of the countries in the sample have developed a market for interest rate swaps in domestic currency. Importantly, in all cases, the counterparty credit risk of these derivatives is very small, either because they are centrally cleared with margin requirements,

or they are bilaterally OTC but they comply with the collateral posting requirements set by the International Swap and Derivatives Association (ISDA)¹⁰. For the safety premium, we use the spread between unsecured interbank loans and the same interest rate swaps as before, of the same maturity. The key here is that the interbank market in these countries operates without collateral requirements, and therefore, they differ in their credit risk.

Proposition 1 shows that standard risk premia might still be present in these spreads (via $l_t^P - l_t^T$), and also its covariance with convenience yields ($\xi_t^{T,d}$)¹¹. In terms of Figure 1, this spread compares a bond close to C and a bond close to A . The spread includes some standard risk premia and a convenience yield in this case. To address this concern, we use assets with short-term maturity for which credit risk is lower (3-month or 1-year maturity). Second, similar to Krishnamurthy and Vissing-Jorgensen (2012), we will show that the effect of proxies of $v'(\theta_t^d/\text{GDP}_t^d)$ will be robust to the inclusion of proxies of risk premia. Appendix C describes the data sources for each country.

2.2 Dollar convenience yield

This measure will capture the convenience yield of the sovereign bond denominated in local currency but in dollar terms. This measure does not capture the convenience yield of the EME sovereign bond denominated in foreign currency.

This can be a relevant measure for an investor who decides on a portfolio of domestic and foreign currency assets and measures her returns in dollars. Appendix shows that, in each country in the sample, a sizable portion of outstanding sovereign bonds denominated in local currency is held by foreigners. As we discuss later in this Section, with the only exception of Turkey, the sovereign bond market in these countries is unlikely to feature market segmentation between currencies and foreign/local investors.

Consider an investor f that measures her returns in dollars and whose portfolio of con-

¹⁰Chung and Chan (2010) show that once ISDA conducted the Credit Support Annex (CSA), which is the general rules for use of collateral, then swap transactions became collateralized and counterparty credit risk seems to be eliminated.

¹¹Since investors will derive no safety benefits from an asset with higher credit or liquidity risk, it is likely that states of the world of higher credit or liquidity risk coincide with states of lower convenience yields. Thus, the covariance between credit risk and the convenience yield will be negative, $\xi_t^{T,d} < 0$.

venience assets is given by:

$$\theta_t^f = \theta_t^{\$M} + \kappa_t^{T,f} \theta_t^T + \kappa_t^{US,f} \theta_t^{US} \quad (5)$$

where $\theta_t^{\$M}$ correspond to dollar money and near-money assets such as U.S. Treasuries; θ_t^T correspond to synthetic dollar bonds: a local-currency sovereign EME bond with all the cash flows swapped into dollars via a forward contract; θ_t^{US} correspond to holdings of non-Treasury safe dollar bond such as a highly rated U.S. corporate bond or a U.S. agency bond. The latter two are of the same maturity. $\kappa_t^{T,f}$ captures the relative convenience service investor f derives from local-currency sovereign bonds of EMEs. As in the previous subsection, time variation in $\kappa_t^{T,f}$ could come from changes in investors' preference for certain assets during, for example, a flight to quality episode.

Consider an investor with investment opportunities in EMEs. An investor with one US dollar in hand today can invest in a highly-rated corporate bond in the US and receive $e^{y_t^{US}}$ at maturity. The investor could instead exchange the US dollar for S_t units of the EME currency and invest in the local-currency sovereign bond to receive $e^{y_t^T}$ units of EME currency at the same maturity. A currency forward contract signed today would convert the EME currency earned into $e^{y_t^T} S_t / F_{t+1}$ dollars. In the ideal world where sovereign bonds are free of default or regulatory risks, and the FX market is frictionless, the two investments would be equivalent, and the spread is equal to zero by no arbitrage.

Financial frictions, convenience yields, default risk, and the possibility of capital controls eliminate riskless arbitrage opportunities, giving rise to profitable trades. For example, if the spread is negative, an investor can earn profits by borrowing at the dollar rate, investing in the EME sovereign bond, and signing a forward contract to convert the EME currency back into dollars¹². Proposition 2 shows how credit and regulatory risks and convenience yields explain the profitability of this trade.

Proposition 2. *The spread between the yield on a non-Treasury safe dollar bond, y_t^{US} , and the yield on the synthetic dollar bond, $y_t^T - \rho_t$, (the yield of the local-currency bond of an EME,*

¹²There is an equivalent interpretation from the perspective of the borrower. The EME sovereign could fund itself at the dollar rate of US corporations and convert the amount raised to the domestic currency in the spot market (hedging with a forward contract) or borrow directly at the domestic rate.

y_t^T , minus the forward premium between the local currency and the dollar, $\rho_t = \log F_{t+1} - \log S_t$, where F_{t+1} and S_t are the forward and spot exchange rates, respectively, both expressed as units of local currency per dollar) can be decomposed as follows:

$$\underbrace{y_t^{US} - (y_t^T - \rho_t)}_{\text{Spread}} = \underbrace{\lambda_t^{T,f} - \lambda_t^{US,f}}_{\text{Diff. Convenience yield}} + \underbrace{(l_t^{US} - l_t^T)}_{\text{Differential default risk}} - \underbrace{k_t}_{\text{Regulatory risk}} + \underbrace{(q_t^T + p_t + (\xi_t^{T,f} - \xi_t^{US,f}) + \psi_t^{T,f})}_{\text{Covariances}} \quad (6)$$

where $\lambda_t^{US,f}$ measures the marginal safety/liquidity services the investor f derives from this US bond (the convenience yield); l_t^{US} is the expected default plus a default risk premium, and $\xi_t^{US,f}$ is the covariance between default risk and the convenience yield. Similarly, $\lambda_t^{T,f}$ is the convenience yield the investor f derives from the sovereign bond of the EME; $l_t^T - q_t$ is the expected loss upon default, l_t^T , net of the covariance between default and currency risk, q_t^T ; $k_t - p_t$ are the expected losses upon the imposition of regulations, k_t , net of the covariance between the risk of regulations and currency risk, p_t ; and $\xi_t^{T,f}$ and $\psi_t^{T,f}$ the covariance of default risk and the convenience yield, and the covariance between the convenience yield and capital control risk, respectively.

If the CIP condition holds for risk-free rates, then $y_{r,f,t} - \rho_t = y_{r,f,t}^{US}$ (later we consider the case where this does not hold). In this case, the dollar convenience yield on local-currency sovereign bonds of EMEs, which is a *differential* convenience yield, corresponds to:

$$\begin{aligned} CY_t^f &\equiv \lambda_t^{T,f} - \lambda_t^{US,f} \\ &= (\kappa_t^{T,f} - \kappa_t^{US,f}) v'(\theta_t^f / \text{GDP}_t^f) \\ &= \left[y_{r,f,t} - \left(y_t^T - l_t^T + q_t^T - k_t + p_t + \xi_t^{T,f} + \psi_t^{T,f} \right) \right] - \left[y_{r,f,t}^{US} - \left(y_t^{US} - l_t^{US} + \xi_t^{US,f} \right) \right] \end{aligned} \quad (7)$$

The second line in Equation (7) gives the model's interpretation of the dollar convenience yield, and the third line gives its empirical counterpart.

In the second line, if EME sovereign bonds share the liquidity services provided by dollar

money, a lower supply of dollar liquid assets such as dollar cash or U.S. Treasuries will increase the dollar convenience yield on EME sovereign bonds through $v'(\theta_t^f/\text{GDP}_t^f)$. In addition, a lower $\kappa_t^{T,f}$ would also reduce the convenience yield. A lower $\kappa_t^{T,f}$ can come from a loss of the safe asset status during a flight-to-quality episode (the dashed line in Figure 1 shifting closer to the straight line) or an increase in default risk (a right-ward move along the horizontal axis in the Figure).

In the third line, the empirical counterpart in Equation (7) shows that $\lambda_t^{T,f}$, in the first square bracket, corresponds to the spread between the local-currency risk-free rate and the sovereign bond. This adjusts the yield on the sovereign bond by all possible risks to replicate a riskless return. In terms of Figure 1, the spread in the square bracket resembles the spread between points B (the risk-free rate) and A . Any spread between the risk-free rate and this riskless sovereign return must come from the safety/liquidity non-pecuniary return, captured by $\lambda_t^{T,f}$. The second square bracket has a similar interpretation for the dollar bond.

Both the default risk and k_t are net of their covariance with currency risk, q_t^T and p_t , respectively. Intuitively, when dollar investors invest in local-currency EME sovereign bonds, default or capital controls cause an additional, indirect loss to them. They not only receive less local currency back, but those cash flows are now worth less if the currency depreciates upon these events. The yield on the synthetic bond does not capture the latter, as currency risk is being hedged. Therefore, the synthetic bond yield underestimates the loss risk upon these events.

The new term $\psi_t^{T,f}$ captures the covariance between the convenience yield and the risk of capital controls. Since higher regulatory risk will lower the non-pecuniary benefits earned by investors, states of the world with higher capital control risk likely coincide with states with lower convenience yields. Thus, the covariance between regulatory risk and the convenience yield will be negative. Similar to the case of $\xi_t^{T,f}$, not accounting for this will overestimate the actual convenience yield of the bond. To correctly capture CY_t^f in the data, we will have to control for measures of credit risk.

How will we recover CY_t^f in the data? Data is available for the spread between the two bonds, $y_t^{US} - (y_t^T - \rho_t)$. Then, we will gather data on differential credit risk (through CDS spreads), regulatory risk plus covariances (explained below), and any residual left will be

attributable to $\lambda_t^{T,f} - \lambda_t^{US,f}$.

There are two main challenges in the identification of CY_t^f . First, how to capture regulatory risk. Second, how to clean for common financial frictions in EMEs and ensure they do not play a role in the residual attributable to $\lambda_t^{T,f} - \lambda_t^{US,f}$. We turn to address both challenges next.

2.2.1 Regulatory Risk

The term k_t captures the risk of regulations imposed by the local government that can inflict additional losses upon investors besides explicit default: taxes on capital outflows, currency convertibility restrictions, and other forms of capital controls. These are relevant as most foreign participation in local-currency sovereign bonds has been through domestic markets under domestic law (Onen, Shin, and von Peter, 2023). The framework above assumes that these regulations impose a loss in the same way as a default would¹³.

The main challenge is finding a proxy for the regulatory risk and the covariances in the right-hand side of Equation (6). We rely on the spread between the swapped local-currency bond and the bond denominated in foreign currency issued offshore. The latter is generally issued under international law and, therefore, is less subject to the unilateral imposition of capital controls and other regulations.

This proxy is based on an analogous exercise carried by Levy Yeyati, Schmukler, and Van Horen (2009) who study the “cross-market premium” between shares of the same companies in EMEs traded simultaneously in the domestic stock market and in New York as depositary receipts (DR)¹⁴. Of course, we use sovereign bonds issued in both markets instead of shares. The authors show that when there are no barriers to cross-country capital movement, arbitrage is expected to equalize the prices of the DR and the underlying share.

¹³These regulations can be imposed and enforced in onshore markets governed by domestic law. However, investors can choose to sign forward contracts offshore that are cash-settled in US dollars and do not require the transfer of the local currency (non-deliverable forwards or NDFs). Therefore, the swapped local currency sovereign bond based on the deliverable currency swap provides the investor with a hedge of the capital control risk and thus should differ from the swapped local currency sovereign bond based on the NDF. Most of the countries in the sample operate NDFs, with the exception of Mexico and South Africa.

¹⁴DRs are certificates traded in major financial centers (New York in this case) that are issued by a depositary bank in the U.S. and represent shares of the ordinary stocks held by a custodian bank in the issuer’s home country. As these underlying stocks can be easily transformed into DRs and vice versa, the stock and the DR represent the same asset traded in two different markets.

In their paper, full integration of capital markets can be disrupted by two main factors: liquidity and capital controls. There might be a liquidity premium that drives transaction costs upwards and reduces the scope for arbitrage¹⁵. In addition, government controls on cross-country capital movement are also expected to affect the cross-market premium¹⁶.

The following Proposition shows that the same should apply to sovereign bonds issued in the same currency but in different markets:

Proposition 3. *Let Φ_t^{FC} denote the spread between the yield of the dollar synthetic bond, $y_t^T - \rho_t$, and the yield of the sovereign bond of the same EME issued offshore in dollars, y_t^{FC} . Then,*

$$\begin{aligned}\Phi_t^{FC} &\equiv y_t^T - \rho_t - y_t^{FC} \\ &\approx (\lambda_t^{FC,f} - \lambda_t^{T,f}) + (l_t^T - l_t^{FC} - q_t^T) + (k_t - p_t) + (\xi_t^{FC,f} - \xi_t^{T,f}) - \psi_t^{T,f}\end{aligned}\tag{8}$$

where $\lambda_t^{FC,f}$ and l_t^{FC} are the convenience yield and the default risk of the sovereign bond issued in dollars, respectively.

The expression Φ_t^{FC} will be approximately equal to the term on regulatory risk plus covariances in (6). However, it also adds two new terms. The first is the differential credit risk between foreign-currency and local-currency bonds. We will assume that $l_t^{FC} \approx l_t^T$. We refer to the discussion by Du and Schreger (2016), where they conclude that recent history in emerging markets does not give a clear higher probability of defaults or higher haircuts in either currency. None of the countries in the sample defaulted during the time covered (starting in December 2007). Before that, only Turkey selectively defaulted on local-currency debt in 1999, and Indonesia selectively defaulted on foreign-currency debt in 2002.

Second, the differential convenience yield between foreign currency bonds and swapped local currency bonds, $\lambda_t^{FC,f} - \lambda_t^{T,f} \equiv \tau_t$. Both convenience yields are in dollars and could differ if the two bonds have different credit risks (already discussed) or different liquidity. As for liquidity, forward contracts used in the swapped local-currency bond have significantly

¹⁵Stocks may not be traded in all markets daily, and stock prices might be sensitive to the trading activity of particular investors because the market is not deep enough.

¹⁶To the extent that these controls are effective in limiting the ability to transfer funds across borders, the cross-market premium would reflect the desire of investors to purchase the securities inside or outside the country.

larger bid-ask spreads and lower trading volume than bonds. Since investors in the swapped local currency bonds have short positions in the less liquid swap market, they have better liquidity overall than holding bonds denominated in foreign currency¹⁷. We will subtract the liquidity risk in currency swaps, measured by their bid-ask spread, from the bond’s convenience yield, to ensure that the dollar convenience yield measure does not include a liquidity premium coming from the forward market¹⁸.

The Internet Appendix IB.1. describes the role of Φ_t^{FC} and shows its evolution in the case of Brazil, which is a country that implemented capital controls during the sample period.

2.2.2 Financial Frictions

Convenience of the dollar currency. The correction using the bid-ask spread of currency swaps also addresses the empirical findings of Jiang, Krishnamurthy, and Lustig (2021), who estimate, for the sample of G10 countries, that most of the convenience of swapped local-currency bonds comes from being swapped into dollars, not from the actual bond. This follows from the liquidity of the dollar currency, which they claim makes any asset denominated in dollars inherit the convenience of the currency. We take the view that, in the case of swapped local currency bonds, their “dollarness” must depend on the liquidity of the forward markets. Suppose a foreigner is investing in a swapped local currency bond. In that case, the liquidity of the dollar currency is well captured by how easy it is to swap the local currency into dollars, which will depend on the liquidity of the EME forward market.

Limits to arbitrage. Engel and Wu (2023) and Du, Im, and Schreger (2018) also consider an additional source of swap market frictions: it could be that the observed forward premium, ρ_t , is different than the hypothetical premium that ensures CIP for risk-free rates. In this case, we would have $\tau_t^{CIP} \equiv y_{rf,t} - \rho_t - y_{rf,t}^{US} > 0$, and the empirical counterpart

¹⁷Du and Schreger (2016), for a sample of 10 EMEs between 2005 and 2014, find that the mean bid-ask spread for local currency debt is 11.1 basis points, and for foreign currency debt is 14.5 basis points. The mean bid-ask spread on five-year currency swaps is 38.2 basis points. Regarding trading volumes, the mean quarterly trading volume is around \$49 billion for local currency bonds, \$25 billion for foreign currency bonds, and \$9 billion for cross-currency swaps.

¹⁸A further friction that could explain this spread are short-selling constraints on swapped local-currency sovereign bonds in EMEs. Du and Schreger (2016) show that lendable inventories and inventory utilization are lower than for foreign-currency debt. Shorting the swapped local-currency bond costs more than the foreign-currency bond (31 basis points against 21 basis points, according to their data). This issue could potentially lower Φ_t^{FC}

in Equation 7 would no longer capture the full dollar convenience yield. Du, Tepper, and Verdelhan (2018) explain that CIP deviations in risk-free rates can arise when the dollar rate is lower than the swapped foreign rate, and banks face balance sheet costs that prevent them from arbitraging the difference. When global financial intermediaries are constrained and demand for dollar liquidity is strong, this shows up as a positive CIP deviation and can lead to the mispricing of forward contracts.

In the case of EMEs, several other regulations can create limits to arbitrage and CIP deviations in risk-free rates. For example, limits on onshore net open positions of forwards, prohibition of resident participation in offshore FX derivatives, tax treatment of non-residents participating in onshore derivatives, or documentation requirements of underlying investments for non-residents¹⁹.

After replacing $\rho_t \equiv y_{rf,t} - \tau_t^{CIP} - y_{rf,t}^{US}$ in the Equations above, the term τ_t^{CIP} shows up twice: in the spread between the dollar bond and the swapped EME bond (in Equation (6) and again in the spread between the swapped EME bond and the foreign currency bond in Equation (8)). As we replace Φ_t^{FC} in (6), the term τ_t^{CIP} cancels out.

However, this correction might not be sufficient, as limits to arbitrage can also arise in bond markets with longer maturities. More generally, recent papers have shown the role of global intermediaries' financial constraints in determining exchange rates (Gabaix and Maggiori, 2015) and the external finance premium (intermediation premium) for EME sovereigns (Morelli, Ottonello, and Perez, 2022). Moreover, financing constraints of domestic financial institutions play a role in the dynamics of short-term local-currency interest rates (De Leo, Gopinath, and Kalemli-Ozcan, 2022).

Therefore, although the mispricing of forward contracts cancels out, it might still be the case that limits to arbitrage in local-currency sovereign bonds or spot markets are driving the spread in (7). Since we do not have direct proxies for these, the empirical results in the following Sections will control for the financial constraints of intermediaries by using capital inflows as a percentage of GDP. The literature has shown theoretically that capital inflows are related to the financial constraints of global intermediaries (Hau and Rey, 2006; Gabaix

¹⁹See Cerutti and Zhou (2024) for an analysis of some of these regulations, the gap between onshore and offshore FX markets in EMEs, and their implication for CIP deviations.

and Maggiori, 2015; Basu et al., 2020).

Bond market segmentation. A further friction in EME bond markets is market segmentation. Foreign investors might only hold dollar assets (non-Treasury safe dollar bonds and EME debt denominated in foreign currency), and local-currency debt might only be held by domestic investors (such as local pension funds). Alternatively, foreign investors might hold local-currency sovereign debt only under foreign jurisdiction (Eurobonds), and domestic investors hold local-currency bonds issued under domestic law. In all these cases, marginal investors for dollar and local currency assets are different, and the spread in (6) captures incomplete arbitrage between the two markets due to arbitrageurs' limited capital and risk aversion.

As mentioned above, Appendix addresses the extent of market segmentation between foreign and local investors, as well as between foreign and domestic currency. In addition, the Internet Appendix IB.2. addresses evidence of the importance of Eurobonds (local-currency sovereign bonds issued under international law and thus not subject to regulatory risk) and how it may affect the estimation in general and market segmentation between international and domestic jurisdictions in particular. Overall, foreign investors are active and relevant holders of local-currency sovereign bonds in all countries in the sample of this paper. The country in the sample most affected by market segmentation is Turkey, and the analysis of the next subsection shows that this translates to a very low dollar convenience yield.

The result in Proposition 4 can be substituted in Equation (6), then move the differential convenience yield to the left-hand side and obtain the following:

$$\lambda_t^{T,f} - \lambda_t^{US,f} = y_t^{US} - (y_t^T - \rho_t) + (l_t^T - l_t^{US}) + \Phi_t^{FC} - \tau_t + (\xi_t^{US,f} - \xi_t^{FC,f}) \quad (9)$$

On the left-hand side, we have the desired convenience yield of EME sovereign bonds against non-Treasury-safe dollar bonds. The Appendix C describes all data sources for bond yields, forward premia, credit risk, and Φ_t^{FC} on the right-hand side of Equation (9). τ_t corresponds to the liquidity risk in forward markets. As for $\xi_t^{US,f} - \xi_t^{FC,f}$, a higher credit risk of the non-Treasury dollar assets or the dollar debt of EMEs is very likely associated

with a lower convenience yield (investors are less willing to pay a premium for their safety). Thus, these covariances have the same sign, and the term is $\xi_t^{US,f} - \xi_t^{FC,f}$ will be small.

What are the non-Treasury dollar bonds, y_t^{US} , in the data? Two series are available—first, the 5-year yield on Resolution Funding Corporation (Refcorp) bonds. As suggested by Longstaff (2004), Refcorp bonds are effectively guaranteed by the U.S. government and are subject to the same taxation, but are not as liquid as Treasuries. The other is the ICE Bank of America index for AAA-rated corporate bonds in the US. It tracks the performance of US dollar-denominated investment-grade-rated corporate debt publicly issued in the US domestic market, although it includes all maturities over one year. How do these dollar assets compare with the local currency sovereign bonds in EMEs in terms of safety and liquidity? The Refcorp and AAA-rated corporate bonds have higher credit ratings than the nine EMEs considered. According to Moody’s, credit ratings are Baa1 on average, ranging from A1 (for Chile) to B3 (for Turkey). Ultimately, since we will be correcting EME yields by their CDS spread, the dollar convenience yield will capture a liquidity premium rather than a safety premium.

We estimate the dollar convenience yield for nine countries: Brazil, Chile, Colombia, Indonesia, Mexico, Peru, the Philippines, South Africa, and Turkey. This selection is based solely on data availability.

3 Local-Currency Convenience Yields in EMEs

In this Section, we first present the summary statistics of each measure. Then, we present further formal evidence that these measures are actually driven by the demand for safety and liquidity services to investors.

3.1 Analysis

Table 1 shows summary statistics for the different measures of the local-currency convenience yield of sovereign bonds. These are calculated on a daily frequency. Columns 1-3 provide moments for the domestic safety premium, measured as the interbank rate on unsecured loans minus the fixed rate on interest rate swaps. Columns 4-6 show the moments of the

Table 1: Summary Statistics

Country	Domestic CY						Dollar CY		
	Safety Premium			Liquidity Premium			(7)	(8)	(9)
	(1) Sample starts	(2) Mean	(3) Std	(4) Sample starts	(5) Mean	(6) Std			
Brazil	n.a.	n.a.	n.a.	Mar. 2007	0.65	22.23	Jun. 2010	28.62	30.31
Chile	May 2010	79.42	40.04	Oct. 2005	-13.36	22.85	Apr. 2011	45.98	26.61
Colombia	n.a.	n.a.	n.a.	Feb. 2011	-27.92	45.36	Dec. 2007	16.76	26.34
Hungary	Jan. 2004	1.20	24.08	Jan. 2004	2.12	29.17	n.a.	n.a.	n.a.
Indonesia	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	Feb. 2015	27.82	15.71
Korea	Jul. 2004	21.11	27.17	Jan. 2004	0.39	11.92	n.a.	n.a.	n.a.
Malaysia	Nov. 2006	14.24	17.97	Nov. 2006	29.14	24.49	n.a.	n.a.	n.a.
Mexico	n.a.	n.a.	n.a.	Jan. 2004	31.45	23.28	Dec. 2007	44.42	24.27
Peru	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	Dec. 2007	29.42	27.00
Philippines	May. 2011	17.08	23.73	May. 2011	2.43	44.63	Dec. 2007	18.40	31.72
Poland	Jan. 2004	11.54	21.50	Jan. 2004	14.78	23.29	n.a.	n.a.	n.a.
South Africa	Jan. 2004	71.44	30.13	Jan. 2004	6.02	60.80	Dec. 2013	27.10	35.57
Thailand	Jan. 2014	55.24	10.92	Jan. 2014	-7.31	11.23	n.a.	n.a.	n.a.
Turkey	Oct. 2006	-45.82	94.00	Feb. 2005	97.87	110.7	Dec. 2007	-4.27	27.11

Notes: Daily frequency. The sample ends on March 9, 2021. The dollar conv. yield uses the yield on Refcorp bonds for y_t^{US} .

liquidity premium of the domestic convenience yield (as in Equation (3)), measured as the fixed rate on an interest rate swap minus the government bond. All of these instruments are denominated in the domestic currency and have a maturity of one year. Columns 7-9 provide moments for the 5-year maturity-dollar convenience yield ($\lambda_t^{T,f} - \lambda_t^{US,f}$ in Equation (9)) using the yield on Refcorp bonds for y_t^{US} .

The magnitude of the liquidity premium is generally very small in Brazil, Chile, Colombia, Korea, Thailand, and South Africa. In these countries, bid-offer spreads are low and market depth in local-currency interest rate swaps is high relative to trading government securities; hence, investors (foreign investors in particular) prefer to manage risks using derivatives as opposed to using cash positions in fixed income markets. Malaysia has only recently taken steps to improve the liquidity of its interest rate swap market. It thus shows a much larger liquidity premium²⁰.

The magnitude of the liquidity premium in Turkey is particularly high. However, as explained in Appendix B, the Turkish financial market is the least developed among the countries in the sample, and there is no evidence of the extent to which that interest rate

²⁰In November 2020, a pilot program was launched to provide flexibility for participating onshore banks and their overseas offices to trade ringgit-denominated IRS with non-resident banks without any underlying commitment to any onshore bank. This flexibility encouraged the participation of non-resident banks in the onshore IRS market, thus promoting liquidity.

swaps are centrally cleared or collateralized; thus, it is very likely that the liquidity premium is contaminated with credit risk or other financial frictions. The interest rate swap for the Philippines uses the Libor rate as the floating leg, so it is more akin to a currency swap. The sample for the Philippines, thus, ends when the Libor rate was discontinued²¹. The empirical results in Section 4 will exclude Turkey and the Philippines from the sample.

For those countries where we can disentangle the domestic convenience yield between safety and liquidity premia, a general pattern emerges, where the safety premium is significantly larger than the liquidity premium (Chile, Korea, the Philippines, Thailand, and South Africa). The European countries in the sample show the opposite pattern (Hungary and Poland).

Figure 2 confirms these insights by showing that the average liquidity premium across countries in the sample is close to zero after the 2008 financial crisis (Appendix D shows the same graph for each individual country). It increased after 2015 when monetary policy in the U.S. and in EMEs became more restrictive, and then dropped sharply in early 2020. Negative values of the liquidity premium should be interpreted as an “inconvenience” yield, which means its liquidity value is lost and falls behind other alternatives.

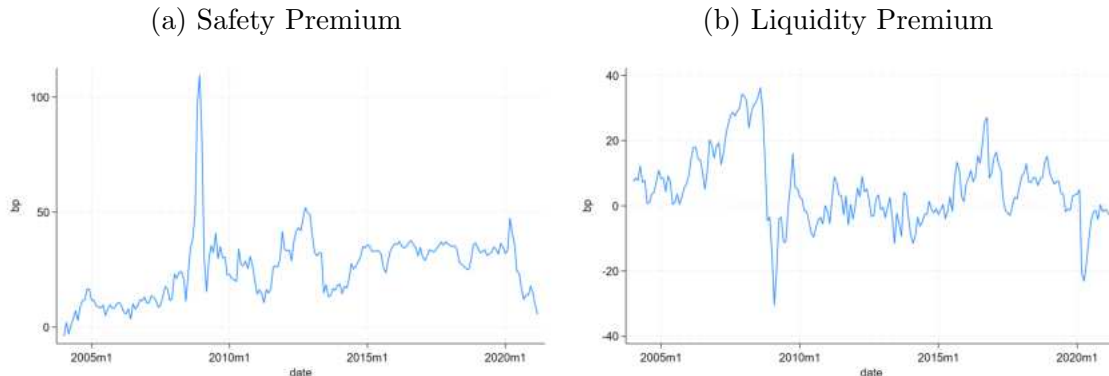
The safety premium, on the other hand, remains sizeable for most of the time span. Interestingly, the safety premium increases around 2012 (during the European debt crisis), and it sharply decreases during the early months of the Covid pandemic.

Column 8 shows that all nine countries for which data are available have a sizable dollar convenience yield. Recall that this is likely capturing a liquidity premium. The only exception is, again, Turkey. This is consistent with the extent of market segmentation. According to BIS data, Turkey has the lowest foreign investors’ participation in local-currency sovereign bonds among these nine EMEs, and it falls below 1% at the end of 2021. Given all these features, it would be worrisome if our measure of dollar convenience yield showed a positive and sizeable magnitude for Turkey.

Figure 3 shows the evolution of the dollar convenience yield computed for the 5-year maturity. A typical pattern emerges: first, an increase around 2011-2012 (that coincides with

²¹The central bank of the Philippines launched a new domestic-currency interest rate swap as recently as 2024.

Figure 2: Domestic Convenience Yield: Safety and Liquidity Premium (1-year Maturity)



Notes: Figure shows the average safety and liquidity premium (as defined in the main text) from Jan. 2004 to Mar. 2021. See Table 1 for the list of countries included.

the Euro debt crisis, a period when some EMEs had lower default risk than some European countries), and another increase starting around 2014-2015 when the Fed started raising rates and dollar liquidity became scarcer. Second, sharp drops during crises, especially the Covid shock in 2020. Significant drops in convenience yields happened in Mexico, Peru, Chile, Indonesia, the Philippines, and South Africa, while Brazil did not experience a significant reduction.

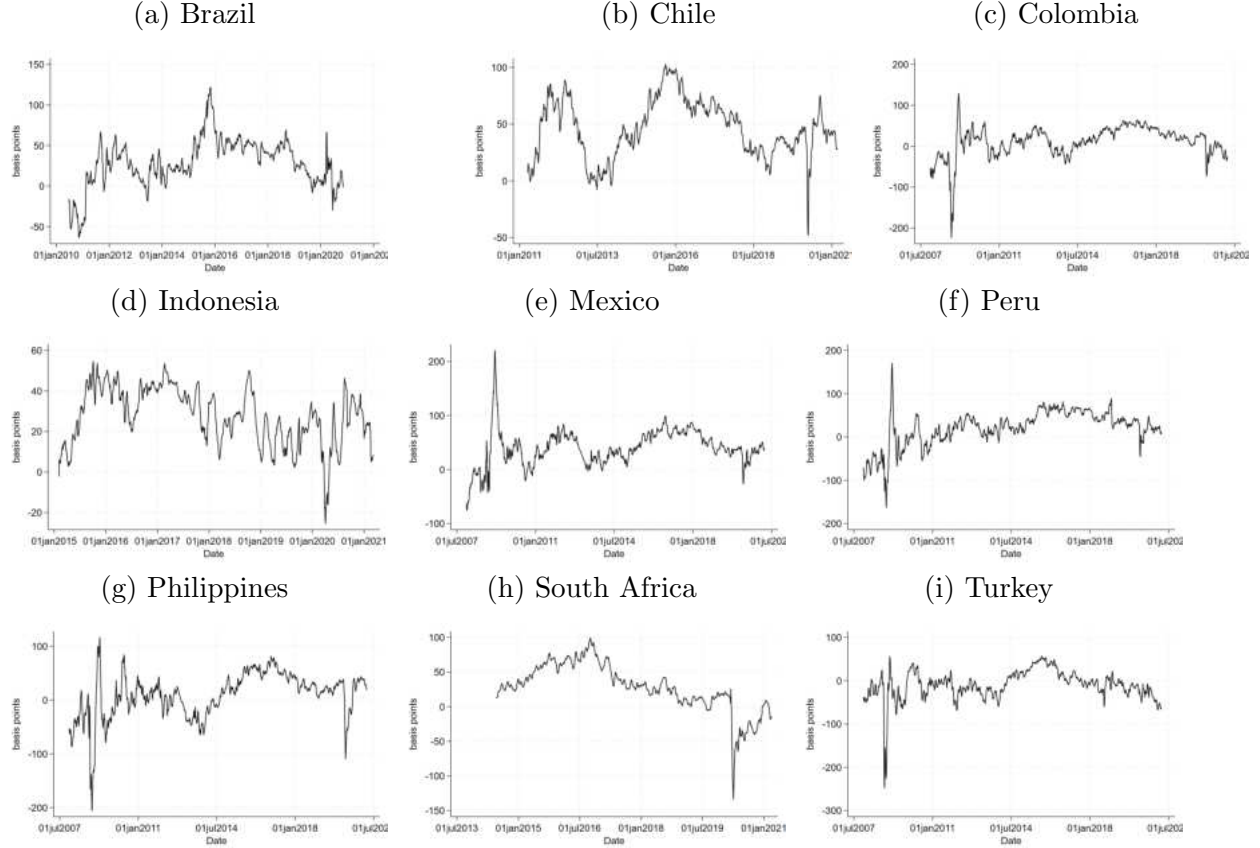
The dollar convenience yield takes negative values at many points in the sample. Recall from Section 2 that this is a relative or differential convenience yield. A negative value should be interpreted as the non-Treasury dollar asset providing more liquidity-related services than the EME sovereign bond. It is consistent, then, that there are significant drops during a flight-to-quality episode, such as the COVID pandemic.

In the Internet Appendix I.B.3., we perform additional analysis on the evolution of the EME convenience yield computed against the U.S. Treasury (the “U.S. Treasury premium” as calculated in Du, Im, and Schreger (2018) for G10 countries).

3.2 The Role of Safety/Liquidity Services

This section provides empirical evidence that the estimated convenience yields capture non-pecuniary services related to safety and liquidity. We follow Krishnamurthy and Vissing-Jorgensen (2012), Greenwood, Hanson, and Stein (2015), and Nagel (2016) and set up the

Figure 3: Dollar Convenience Yield on 5-Year Local-Currency Sovereign Bonds



Notes: Figure shows each country's 14-day moving average of the dollar-convenience yield.

following regression:

$$cy_{i,t} = \beta_1(\text{Gov. debt supply/GDP})_{t-1} + \beta_2 i_{t-1}^{MP} + \beta_3 i_{t-1}^{US} + \beta_4 X_t + c_i + \tau_t + \epsilon_{i,t} \quad (10)$$

where i is currency/country, t is time (monthly), and $cy_{i,t}$ is either the dollar or the domestic convenience yield. The variable $(\text{Gov. debt supply/GDP})_t$ is the outstanding supply of “safe assets”. This is proxied by either the local-currency sovereign debt or the U.S. government debt supply. Both quantities are net of central bank holdings. For the domestic convenience yield, this is a proxy for θ_t^T/GDP_t^d in the model's Equations (2) and (4), and for the dollar convenience yield, a proxy for θ_t^{SM}/GDP_t^f in (5) and (7). If investors have a special demand for safety and liquidity, the coefficient β_1 represents the slope of the demand curve for safe assets and should, therefore, be negative. If there is no special demand for safety and liquidity, then we are back to standard asset pricing. The spread $cy_{i,t}$ would

capture only a standard risk premium, and the supply of debt should play no significant role ($\beta_1 = 0$).

The variables i_t^{MP} and i_t^{US} correspond to the level of the monetary policy rate in each EME and the U.S., respectively. As explained in Nagel (2016) and Diamond and Van Tassel (2023), these proxy the price of the most liquid asset in the economy: money or its near substitutes, such as central bank reserves or private liquid deposits. If government debt shares the money properties of very liquid assets, then its convenience yield should respond positively to the price of money, i.e., the level of the monetary policy rate. In terms of the model above, this is a proxy for θ_t^M and $\theta_t^{\$M}$. Higher levels of interest rates are associated with a lower supply of money assets, driving up their convenience yield.

Lastly, X_t refers to relevant control variables. According to Proposition 1, the measure of the domestic convenience yield ($y_t^P - y_t^T$) might include a credit risk component. Therefore, we control for proxies of default risk and risk aversion to show that they do not drive the results. For the dollar convenience yield, these controls are crucial to ensure that the residual differential convenience yield does not capture any covariance between the convenience yield and credit risk.

The independent variables are lagged one month to avoid endogeneity and reverse causality as much as possible. The variables c_i and τ_t are country and time-fixed effects, respectively. Year-fixed effects capture time-varying unobserved variables common to all countries. In this sense, it controls for global variables other than the supply of U.S. debt and the federal funds rate level. We double-clustered the standard errors across year and country.

Table 2 presents the results for the domestic and dollar convenience yields. To save space, the domestic convenience yield combines the safety and liquidity premiums. The Internet Appendix I.C1 presents the same regression separately for the safety and liquidity premiums.

Columns 1 and 4 show that both measures respond positively to the level of the monetary policy rate. As explained above, this is the sign one expects if EME local-currency sovereign bonds have some degree of substitution with money, which is the safest and liquid asset. A higher monetary policy rate is associated with a lower supply of money-related assets, which increases the convenience yield on other near-money assets, such as government debt. The local monetary policy rate has a more significant effect on the domestic convenience yield.

Table 2: Determinants of Convenience Yields

	Dep. var.: Domestic CY			Dep. var.: Dollar CY		
	(1)	(2)	(3)	(4)	(5)	(6)
Local MP rate _{t-1}	7.988*** (2.082)	13.41*** (2.324)	7.991*** (2.078)	0.350 (0.547)	0.083 (0.739)	0.343 (0.544)
U.S. MP rate _{t-1}	-0.990 (4.925)	-1.065 (4.816)	-0.698 (3.810)	11.58*** (3.817)	9.131** (3.857)	10.90*** (4.012)
$\log(\frac{\text{Local gov debt}}{GDP_{local}})_{t-1}$	-31.64*** (6.407)	-31.39*** (6.782)	-31.64*** (6.408)	8.27 (8.510)	2.89 (8.288)	8.219 (8.51)
$\log(\frac{\text{U.S. gov debt}}{GDP_{US}})_{t-1}$	45.80 (52.44)	48.70 (51.94)	46.31 (55.21)	-135.9*** (45.04)	-132.0*** (43.63)	-137.2*** (46.86)
slope _{local,t-1}		11.15*** (1.710)			-0.107 (1.550)	
slope _{US,t-1}			0.499 (5.449)			-1.378 (4.019)
Constant	-37.01 (107.1)	-53.24 (106.6)	-37.07 (106.9)	-345.1*** (82.13)	-338.9*** (80.29)	-345.3*** (83.1)
Observations	1,149	1,149	1,149	1,137	1,103	1,137
R-squared	0.500	0.521	0.500	0.660	0.676	0.663

Notes: Data are at monthly frequency. All columns include country and year-fixed effects. The domestic conv. yield is the sum of the safety and the liquidity premium. The dollar conv. yield uses the yield on Refcorp bonds for y_t^{US} . Standard errors are double-clustered by country and year. Start dates vary among countries but end in March 2021 for all. U.S. debt and EME local-currency debt-to-GDP variables are net of the central bank's holdings. *** p<0.01, ** p<0.05, * p<0.1

In contrast, the U.S. monetary policy rate has a significant impact on the dollar convenience yield. The magnitudes of the coefficients are in line with those in Nagel (2016) for the US.

Regarding the supply variable, in Columns 1 and 4, the supply of government debt has a negative impact on the convenience yield. The supply of local-currency bonds has a significant adverse effect on the domestic convenience yield, and the supply of U.S. government debt decreases the dollar convenience yield. This suggests that both measures of convenience yields correctly capture the relevant currencies for each investor: a larger supply of local-currency government bonds affects more the convenience yield that measures returns in the local currency and analogously for the dollar measure.

The negative coefficient on the relative supply of U.S. Treasuries is a crucial result. As explained above, if the measures of local-currency convenience yield capture demand for safety and liquidity, then the estimated coefficient represents the slope of the demand for safe assets and, therefore, should be negative. Under a standard asset pricing model, standard

risk premium (differences in credit risk between two assets, as illustrated by points C and D in Figure 1) does not depend on the supply of the asset. The negative coefficient indicates that what is captured is a convenience yield, representing a deviation from the standard asset pricing model.

Columns 2 and 5 show that results are robust to the inclusion of a proxy for risk premia (the yield curve’s slope in domestic currency). Columns 3 and 6 use the slope of the dollar yield curve. The slope of the yield curve is known to predict the excess returns on stocks, and it is a commonly used risk factor when estimating risk premia in bond markets (Campbell and Shiller, 1991; see Baumeister (2023) for a comprehensive review). For example, investors who are more risk-averse in a recession will demand a higher risk premium to hold the sovereign bond or its private substitutes. Thus, the yield curve’s slope serves as a measure of variation in the risk premium component of the bond spread. In addition, to the extent that default and liquidity risk are likely to vary with the business cycle, the slope variable can also help control for the expected risks in the yield spread.

We measure the slope as the spread between the 10-year sovereign bond yield and the 3-month yield. The estimated coefficients for the supply of debt and the monetary policy rate remain robust even with the inclusion of the slope variable. This suggests that the results are not driven by the standard risk premia that investors charge on EME debt, and that the spreads are driven mainly by $v'(\cdot)$ in the model.

The Internet Appendix I.C1 shows that the regression results still hold when we disentangle the domestic convenience yield into its safety and liquidity components. In addition, when we add the 1-year CDS spread to the regression, it has a significant but very small effect on the safety premium, and the other coefficients still hold. It also replicates the dollar convenience yield regression using the ICE index for AAA-rated corporate US bonds for the yield y_t^{US} and shows that results generally hold²². In the Internet Appendix I.C2, we run the regression (10) but with credit risk (measured as the differential CDS spread) as a dependent variable. If our decomposition accurately disentangled local-currency bond premiums from default risk, then the determinants should differ. In contrast to convenience yields, credit

²²The issue with this measure is that the index includes corporate bonds of all maturities longer than one year, so in this case, the convenience yield includes some standard term premia

risk does not respond to the debt supply, which is what standard asset pricing predicts for risk premia.

3.3 Discussion

How do these convenience yields relate to recent findings of demand-based asset pricing for EME sovereign bonds and the role of heterogeneous investors?

Fang, Hardy, and Lewis (2022) analyze investor demand for sovereign debt using a demand system approach based on low-frequency country-level data of sovereign debt ownership split by banks and non-banks, both foreign and domestic. They find that private non-banks absorb most new debt issuance and are the creditor group most responsive to the yield. Zhou (2024) focuses on a more detailed foreign investor split: investment funds (prone to risk-sensitive redemptions) and banks, insurers, and pension funds with a more stable demand structure. He finds that foreign insurers and pension funds tilt their emerging market portfolio towards securities with higher credit quality, and their sensitivity to the shifts in the VIX index is lower than for foreign investment funds. Moreover, during the Taper Tantrum and the Covid pandemic, foreign banks, insurers, and pension funds responded by buying EME sovereign debt, while investment funds became net sellers. In the same line, Converse, Levy-Yeyati, and Williams (2023) show that exchange-traded funds (ETFs) amplify EMEs' sensitivity to the global financial cycle.

According to these findings, it is plausible that foreign insurers and pension funds could be driving the dollar convenience yield estimated in Section 2. Their demand structure is more stable, they are less sensitive to global risk factors, they have a downward-sloping demand curve for EME sovereign debt, and they do not face the strong redemption pressure investment funds face during episodes of heightened global uncertainty.

Moretti et al. (2024) present evidence of downward-sloping demand curves for risky sovereign bonds, which works which discourage the government from borrowing too much. Their paper features a structural model with a demand structure that includes active and passive investors. The equilibrium bond price includes a function that captures the demand's downward-sloping nature and which they identify with a "convenience yield". However, by construction, this function decreases the bond price, so they assume an "inconvenience yield".

Within their paper, this is consistent because they calibrate the model to Argentina. While their notion of “convenience yields” refers to anything that makes demand inelastic, the approach by Krishnamurthy and Vissing-Jorgensen (2012) and Du, Im, and Schreger (2018) can be seen as a price-based way to link a spread between assets to a demand for safety and liquidity more specifically.

4 Sovereign Bonds as Safe Havens

What can convenience yields tell us about the safe-haven status of government bonds? Appendix E proposes a model that links the comovement of convenience yields with global and country-specific systematic risk to their safe-haven status.

In particular, an asset is a safe haven against global risk if its safety and/or liquidity value (measured by the convenience yield) increases with the global risk factor. Similarly, an asset is a safe haven against local or country-specific risk if its safety or liquidity value (measured by the convenience yield) increases with the local risk factor.

The model in Appendix E rationalizes the following empirical regression of convenience yields on global and local risk factors:

$$CY_t^j = \gamma_1 \rho_t^{US} + \gamma_2 \rho_t^{EM} + c_i + \tau_t + \epsilon_{i,t} \quad (11)$$

for $j \in \{d, f\}$. i is currency/country, t is month, $cy_{i,t}$ is the convenience yields, and the independent variables are lagged one month. The regression also includes country-fixed effects, and standard errors are double-clustered across month and country. A $\hat{\gamma}_1 > 0$ means the sovereign bond is a safe haven against global risk. Similarly, a $\hat{\gamma}_2 > 0$ means the sovereign bond is a safe haven against country-specific risk.

The global factor, ρ_t^{US} , captures investors’ risk sentiment on the global economy (Rey, 2013; Miranda-Agrippino and Rey, 2022). This can also relate to financial frictions on intermediaries that limit their ability to engage in arbitrage (Gabaix and Maggiori, 2015; Morelli, Ottonello, and Perez, 2022). Below, we control for the latter to isolate the effect of investors’ risk sentiment. The local factor, ρ_t^{EM} , analogously captures investors’ risk

sentiment towards a given country or country-specific frictions that can arise from economic policy uncertainty affecting investors.

To capture global risk sentiment, we utilize the VIX and the global risk factor introduced by Miranda-Agrippino and Rey (2020, 2022). This factor is built on global asset prices, including equity prices²³, commodity prices excluding precious metals, and corporate bond indices. This global factor accounts for approximately a quarter of the variance in risky asset prices and about 35% of the variance in gross capital flows.

How important are global factors for EMEs? González-Rozada and Levy Yeyati (2008) find that the evolution of global factors such as risk appetite and global liquidity explains a large fraction of the time variability of EME bond spreads. In particular, they find that international risk appetite and international liquidity are two exogenous risk factors that explain over 50% of the long-run time variability of EME spreads. By contrast, debt-specific factors, such as credit ratings or outlook changes, lag spread movements and elicit little to no additional effect on spreads.

We employ the Economic Policy Uncertainty (EPU) index, developed by Baker, Bloom, and Davies (2016), to capture local risk sentiment. The index is available for Brazil, Chile, Colombia, and Mexico. To complement and as a robustness check, we also employ the risk index from the International Country Risk Guide (ICRG) dataset, which provides data on a country’s political, economic, and financial risks for more than 140 countries at a monthly frequency.

How important are these variables for EMEs? The selection of these variables is based on the findings of Kalemli-Ozcan and Varela (2023), who show that these capture a relevant factor explaining country-specific systematic risk in emerging market economies. In particular, country-specific policy uncertainty differs from default risk, as it stems from volatility in a broad set of policies, including monetary policy uncertainty, fiscal policy uncertainty, expropriation risk, and democratization risk.

The EPU index is built using monthly counts of articles in local newspapers that convey the extent of uncertainty. In the case of the indices for Brazil, Chile, Colombia, and Mexico, the keywords an article must contain to be considered in the index are “uncertainty” and

²³Covering Europe, North America, Latin America, Asia Pacific, and Australia

“economy/economic”²⁴.

The ICRG gives a score for each source of risk in a country. We will include the financial and economic risk variables, each of which has five components, and their assessments are based solely on objective data. Economic risk encompasses GDP per capita, real GDP growth, inflation rate, budget balance as a percentage of GDP, and current account balance as a percentage of GDP. Financial risk includes foreign debt to GDP, foreign debt service over exports of goods and services, current account over exports of goods and services, net international liquidity as months of import cover, and exchange rate stability. Notice that, while these two variables are country-specific, a broad distinction can still be made between them. Economic risk captures the country fundamentals and the business cycle. Financial risk, on the other hand, is related to government debt service and the liquidity available to governments, especially in relation to foreign debt. As we shall see below, the financial risk will have a different effect on the safety premium.

Table 3 shows the results for the domestic convenience yield. Recall that for the remainder of this Section, regressions exclude Turkey and the Philippines. *mar* is the global risk factor in Miranda-Agrippino and Rey (2020). The risk variables from the ICRG are standardized by each country’s mean and standard deviation (higher values reflect higher risk). Panel A shows the results for the safety premium. It does not include the EPU as an independent variable because there is only one country with data for the safety premium and the EPU (Chile). Panel B shows the results for the liquidity premium, for which there are more countries with data available (see Table 1).

The domestic safety premium correlates positively with both the global risk factor and the country-specific factor that measures economic risk. This suggests that, compared to private domestic assets, the local currency sovereign bond works as a safe haven for both types of risks. The negative effect of the financial risk index can stem from its more direct relationship to government debt service capacity (foreign debt to GDP and foreign debt service as a percentage of exports of goods and services) and the liquidity available to governments (net

²⁴After obtaining a raw count, the number of articles is scaled by the total number of articles in the same newspaper and month. Then, they standardize each newspaper’s scaled frequency counts to have a unit standard deviation within a period of time (usually starting in the mid-1990s or early 2000s). Finally, they normalize the series to have a mean of 100 over the same period.

Table 3: Domestic Convenience Yield and Risk Factors

Dep. var.: dom. CY	(1)	(2)	(3)	(4)	(5)	(6)
A. Safety Premium						
VIX_{t-1}	1.388*** (0.273)		1.374*** (0.313)			
mar_{t-1}		16.33*** (3.808)		16.09*** (4.273)		
Economic risk $_{t-1}$	7.713*** (1.898)	7.706*** (1.922)				
Financial risk $_{t-1}$			-2.977** (1.188)	-2.918** (1.195)		
Observations	719	719	719	719		
R-squared	0.779	0.777	0.767	0.764		
B. Liquidity Premium						
VIX_{t-1}	-1.000*** (0.247)		-0.712** (0.323)		-0.717** (0.332)	
mar_{t-1}		-12.67*** (2.653)		-10.44*** (3.886)		-10.90*** (3.902)
EPU_{t-1}	13.31*** (2.229)	12.59*** (2.151)				
Economic risk $_{t-1}$			-2.502 (2.589)	-2.578 (2.670)		
Financial risk $_{t-1}$					3.154** (1.509)	3.781** (1.575)
Observations	595	595	1,153	1,153	1,153	1,153
R-squared	0.574	0.568	0.303	0.306	0.307	0.309
Country & Year FE	Y	Y	Y	Y	Y	Y
MP rate & Debt supply	Y	Y	Y	Y	Y	Y
K inflows	Y	Y	Y	Y	Y	Y

Notes: *mar* is the global risk factor in Miranda-Agrippino and Rey (2020). *EPU* is the economic policy uncertainty index of Baker, Bloom, and Davis (2016) -available for fewer countries. Regressions exclude data for Turkey and the Philippines. Data are at a monthly frequency. Standard errors are double-clustered by country and month. Start dates vary among countries, but ICRG data is available until April 2019. *** p<0.01, ** p<0.05, * p<0.1

international liquidity, expressed as months of import cover). Not surprisingly, the safety premium declines with these risk factors.

The domestic liquidity premium exhibits varying responses depending on the nature of the risk. The liquidity value significantly decreases with global risk, but it increases with local risk. Regarding the model in Appendix E, this suggests that in times of higher global risk, investors do not perceive the local currency sovereign bond as retaining its value and will find it more difficult to sell than alternatives.

To further address the concern about the presence of standard risk premia in these spreads, the Internet Appendix IC.3. replicates Table 3 for the 3-month maturity, where

Table 4: Dollar Convenience Yield and Risk Factors

Dep. var.: dollar CY	(1)	(2)	(3)	(4)	(5)	(6)
VIX_{t-1}	-0.828*** (0.306)		-0.939*** (0.283)		-1.061*** (0.264)	
mar_{t-1}		-21.32* (12.20)		-15.03 (11.75)		-22.81** (11.05)
EPU_{t-1}	13.27*** (2.478)	12.33*** (2.253)				
Economic risk $_{t-1}$			2.195 (2.155)	3.254 (2.182)		
Financial risk $_{t-1}$					5.610*** (2.013)	4.538** (2.264)
Country & Year FE	Y	Y	Y	Y	Y	Y
MP rate & Debt supply	Y	Y	Y	Y	Y	Y
K Inflows	Y	Y	Y	Y	Y	Y
Observations	540	540	873	873	873	873
R-squared	0.738	0.719	0.706	0.695	0.709	0.700

Notes: *mar* is the global risk factor in Miranda-Agrippino and Rey (2020). *EPU* is the economic policy uncertainty index of Baker, Bloom, and Davis (2016) -available for fewer countries. Regressions exclude data for Turkey. Data are at monthly frequency. Standard errors are double-clustered by country and month. Start dates vary among countries but ICRG data is available until April 2019. Capital inflows-to-GDP variables are standardized by the mean and standard deviation of each country. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

arguably risk premia is very small. In the case of the US, it is well known that short-term T-bills are the most money-like in the term structure (Greenwood, Hanson, and Stein, 2015; Du, Im, and Schreger, 2018). Results hold, although with different magnitudes. Notably, the liquidity premium at the 3-month maturity responds positively to global risk, which suggests the same mentioned pattern than in the US.

Table 4 shows the results for the dollar convenience yield. Overall, the picture is similar to the domestic liquidity premium. While the dollar convenience yield responds positively to local risk, it responds strongly negatively to global risk.

Notably, this suggests that the local currency sovereign bond serves as a safe haven against local or country-specific risk, compared to a foreign asset. At the same time, it does not work as a safe haven against global uncertainty.

All columns control for the responses of the monetary policy rate and the supply of government debt. This indicates that results are not driven by the responses of the central bank and the fiscal authority, which could alter the supply of safe and liquid assets in response to varying levels of uncertainty.

In addition, all columns add capital inflows over GDP as controls. As explained above, the global factor can capture both the role of investors’ risk appetite but also the role of financial constraints of global intermediaries (Gabaix and Maggiori, 2015). In the case of EMEs, these constraints can lead to an intermediation premium on sovereign debt (Morelli, Ottonello, and Perez, 2022). Capital inflows work as a proxy for the latter because they have been shown to strongly correlate with financial constraints of global intermediaries (Hau and Rey, 2006; Gabaix and Maggiori, 2015; Basu et al., 2020). If the coefficient of the global factor remains significant after controlling for capital inflows, this would confirm that global uncertainty has a negative effect on its own. We added capital inflows disaggregated by the sector they are directed to (government, bank, or corporate debt), using data from Adjiev et al. (2022).

Combined, Tables 3 and 4 suggest that the safe asset status of local currency sovereign bonds in EMEs depends on the nature of the shock and the benchmark asset. In addition, these results suggest that the commonly held view that capital inflows are primarily driven by yield-oriented investors may not always be accurate.

If we assume that the domestic convenience yield originates from a domestic investor and the dollar convenience yield from a foreign investor, then these results have significant implications for the investor base in EMEs. While domestic investors consider the local currency sovereign bond a safe asset for local and global risk, foreign investors only do so for local shocks. This can add an extra source of volatility during episodes of global uncertainty.

4.1 Analysis of Two Exogenous Shocks

This subsection aims to better understand why the dollar convenience yield declines in the face of higher global uncertainty. Within the framework of Section 2, there are two major reasons why the convenience yield could drop. One is that the credit or liquidity risk increases, which mechanically reduces the convenience yield. Regarding Figure 1, we move right along the horizontal axis, and investors are less willing to pay a safety/liquidity premium for this asset. The other is a loss of the safe asset status or a drop in $\kappa_t^{T,f}$ (this moves the dashed line closer to the straight line). This can be due to repricing amid higher uncertainty or coordination into a new equilibrium with a different safe asset.

We test this by analyzing the response to two identifiable exogenous shocks to EMEs: the Taper Tantrum and the Covid pandemic. These are widely accepted as exogenous and unanticipated adverse shocks to EMEs. The Taper Tantrum started with Fed Chairman Ben Bernanke’s speech in May 2013, which signaled the end of the Fed’s large-scale asset purchases and, thus, a future reduction in the supply of dollar liquidity. The Covid-19 episode likely represents many shocks; therefore, we will focus our analysis on the early months of the pandemic (March-June 2020). Both episodes involved increased risk and a capital inflow reversal for EMEs. Still, one difference is that, unlike the first months of the Covid shock, the Taper Tantrum did not trigger a flight to safety episode (understood as global investors buying U.S. Treasuries due to their perceived safety). This can be seen in the response of the VIX index (which did not spike).

We run a regression with the dollar convenience yield on the left-hand side and interact the shocks with the explanatory variables of the previous sections. The interacted coefficients capture any change in the sensitivity of convenience yields to the different determinants and will shed light on which variables most likely drive the responses during these episodes. Results show that the response of the dollar convenience yield is quite different in the two episodes, driven in each case by different explanatory variables. Compared to Table ??, we introduce the supply of debt as the relative supply of U.S. Treasuries over local debt, which allows for a less noisy estimation of the supply coefficient.

Table 5 shows the results. Column 1 shows that the Taper Tantrum had a positive and significant effect. In Column 2, the coefficient of the interaction between the shock and the local monetary policy rate is positive and statistically significant—this variable proxies for the price of money and near-money assets. As explained before, a higher monetary policy rate is associated with a higher price of liquidity. The positive sign of the interaction term suggests that the convenience yield increased due to the liquidity shortage during the episode. Recall that during the Taper Tantrum, there was no flight to safety but scarcer liquidity that plausibly drove up the convenience yield of sovereign bonds. As Column 2 shows, this effect is not driven by the rise in risk premia, as captured by the slope of the local yield curve.

In contrast, Column 3 shows that the Covid shock significantly reduced the convenience yield by almost 19 basis points. In Column 4, the interaction of the shock with the relative

Table 5: Effect of Taper Tantrum and Covid-19 Shocks

Dep. var: $cy_{i,t}$	(1)	(2)	(3)	(4)
Non-interacted regressors	Yes	Yes	Yes	Yes
TT_{t-1}	4.875*** (1.348)	2.972 (2.778)		
$MP\ rate_{t-1} \times TT$		2.030*** (0.524)		
$\log(\frac{US\ debt\ to\ GDP}{Local\ Debt\ to\ GDP})_{t-1} \times TT$		0.682* (0.366)		
$vix_{t-1} \times TT$		-0.783** (0.379)		
$slope_{local,t-1} \times TT$		0.413 (1.551)		
$Covid-19_{t-1}$			-18.92*** (5.908)	-21.84*** (5.517)
$MP\ rate_{t-1} \times Covid-19$				-1.830 (1.513)
$\log(\frac{US\ debt\ to\ GDP}{Debt\ to\ GDP})_{t-1} \times Covid-19$				-2.358*** (0.601)
$vix_{t-1} \times Covid-19$				0.570* (0.288)
$slope_{local,t-1} \times Covid-19$				3.108 (1.930)
Constant	46.92** (18.32)	49.41*** (18.55)	47.11** (18.60)	51.79** (19.70)
Year FE	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes
Lagged dep. var.	Yes	Yes	Yes	Yes
Observations	1,091	1,091	1,091	1,091
R-squared	0.838	0.839	0.841	0.846

Notes: See Table ?? . TT is a dummy variable taking the value one from May to December 2013. Covid-19 is a dummy variable taking the value one from March to June 2020. All columns include country and year fixed effects *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

supply of U.S. Treasuries is significantly negative, suggesting that the demand for this global safe asset has become significantly steeper. This is consistent with a global flight to the safety of U.S. Treasuries and with global investors preferring this safe asset over the local sovereign bond.

Column 4 suggests that what drives the drop in convenience yield is not the mechanical rise in credit risk or risk premia charged by global investors during this type of episode, but the availability and move towards alternative global safe assets. This further demonstrates that demand for safety is a significant driver of capital flows for both advanced and emerging economies.

5 Conclusion

This paper shows that convenience yields due to safety and/or liquidity services are relevant in pricing local-currency sovereign bonds in EMEs. However, this does not make them equivalent to a U.S. Treasury or an advanced economy. The responses of the convenience yield against global and local risk factors suggest that their safe asset status depends on the nature of the shock and the benchmark asset. They are a safe haven compared to domestic private assets against country-specific and global risks. However, they are a safe haven compared to foreign private assets only against country-specific risk. Evidence from the Taper Tantrum and the Covid episodes suggests that the explanation does not rest on higher credit risk or risk premia, as expected, but on losing the safe asset status due to a shift in preferences towards U.S. Treasuries. This further suggests that demand for safety is a relevant driver of capital flows for advanced and emerging economies.

The dynamics of convenience yields and their response to global shocks have important implications for EMEs that call for more research on this topic. For example, losing the safe asset status could lead to higher interest rate volatility, which can impact the fiscal capacity of EME governments. Additionally, convenience yields are one reason for bond demand to be downward sloping, allowing large-scale asset purchases (LSAPs) by EMEs' central banks to impact yields. Indeed, many EME central banks conducted such purchases during the Covid crisis. Therefore, if the convenience yield from global investors drops during a crisis, that will limit the effectiveness of LSAPs.

Finally, given the relevance of the demand for safety and liquidity, more research is needed to explain the cross-sectional differences in convenience yields among different types of countries (primary surpluses, low inflation risk, regulations, etc.). In addition, future research can aim to better understand how standard policies for EMEs interact with convenience yields, such as reserve accumulation, different forms of capital controls, or foreign exchange intervention.

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Appendix A Holders of Local-Currency Sovereign Bonds

This Section shows that, for all the countries included in the sample, the vast majority of the outstanding debt is denominated in their local currency. This speaks to the relevance of the question of whether this debt serves as a safe asset or not.

Second, it shows that, in all countries except Turkey, a sizable portion of this local-currency sovereign debt is held by foreigners. Therefore, the dollar convenience yield is unlikely to capture market segmentation and limits to arbitrage between foreign and local markets.

The data presented here comes from Onen et al. (2023), who study the evolution of the currency denomination of sovereign debt in a large set of emerging economies. The dataset covers sovereign bonds, defined as long-term debt securities with original maturities exceeding one year, issued by general governments in both domestic and international markets. The general government comprises central, state, and local government and social security funds, but excludes state-owned companies and the central bank. Outstanding amounts and foreign holdings are in nominal value where available, and market values otherwise, and expressed in current US dollars at end-of-period exchange rates.

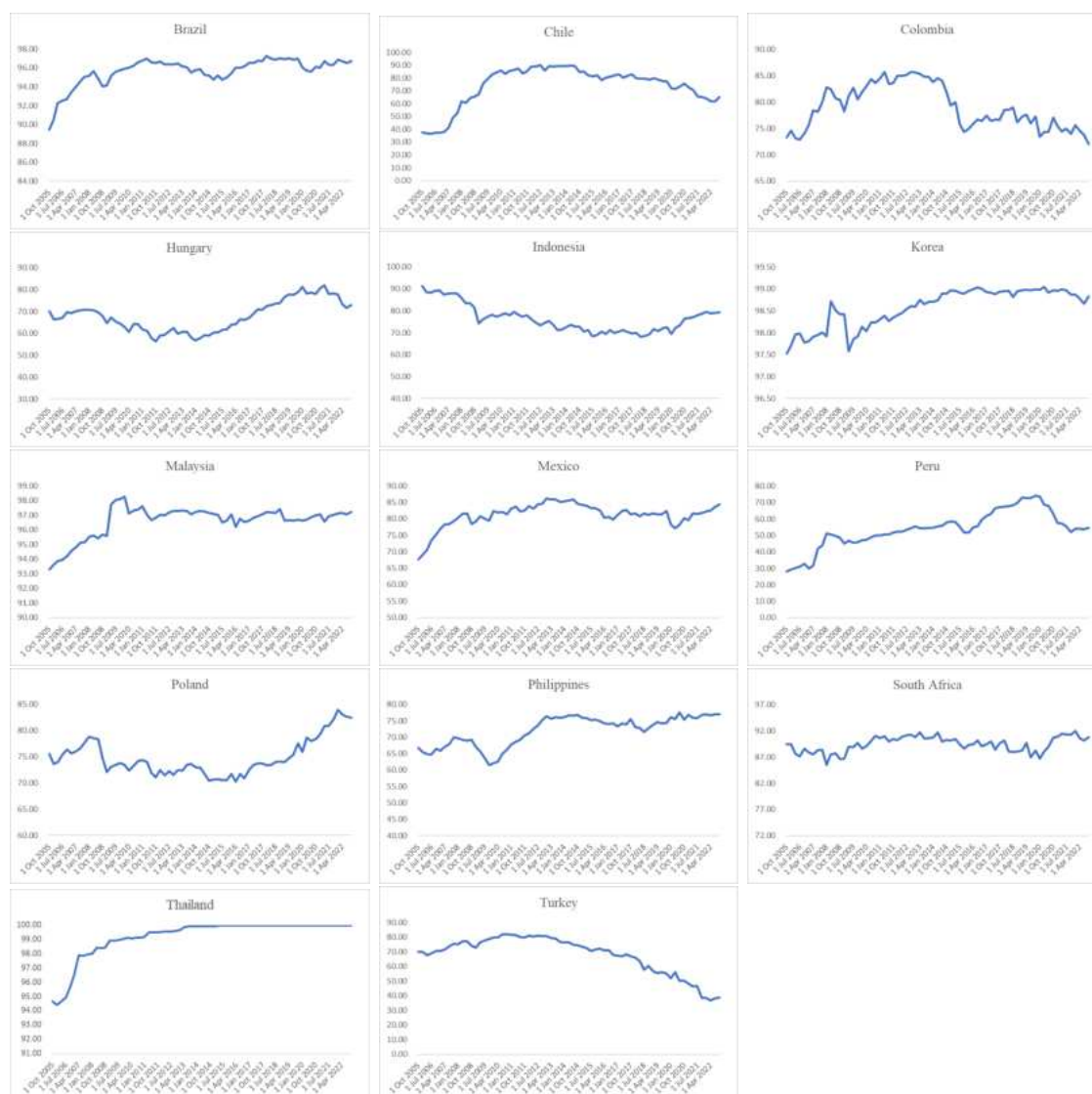
Figure 4 shows, for each country, the share of local currency in government bonds (as % of total outstanding debt in all currencies).

Figure 5 shows, for each country, the foreign investor share in local-currency bond market (as % of local-currency bonds outstanding). Thus, it shows the percentage of all sovereign bonds issued in domestic currency held by foreign investors.

Appendix B Financial Markets By Country

We provide country-by-country evidence on the (1) central role played by domestic currency government bonds in the local financial market, (2) extent to which the interbank markets rely on unsecured and illiquid loans, and (3) extent to which domestic currency interest rate swaps carry low credit risk. Sources are the various documents from the Financial Sector Assessment Program by the IMF and the World Bank, internal documents by local central

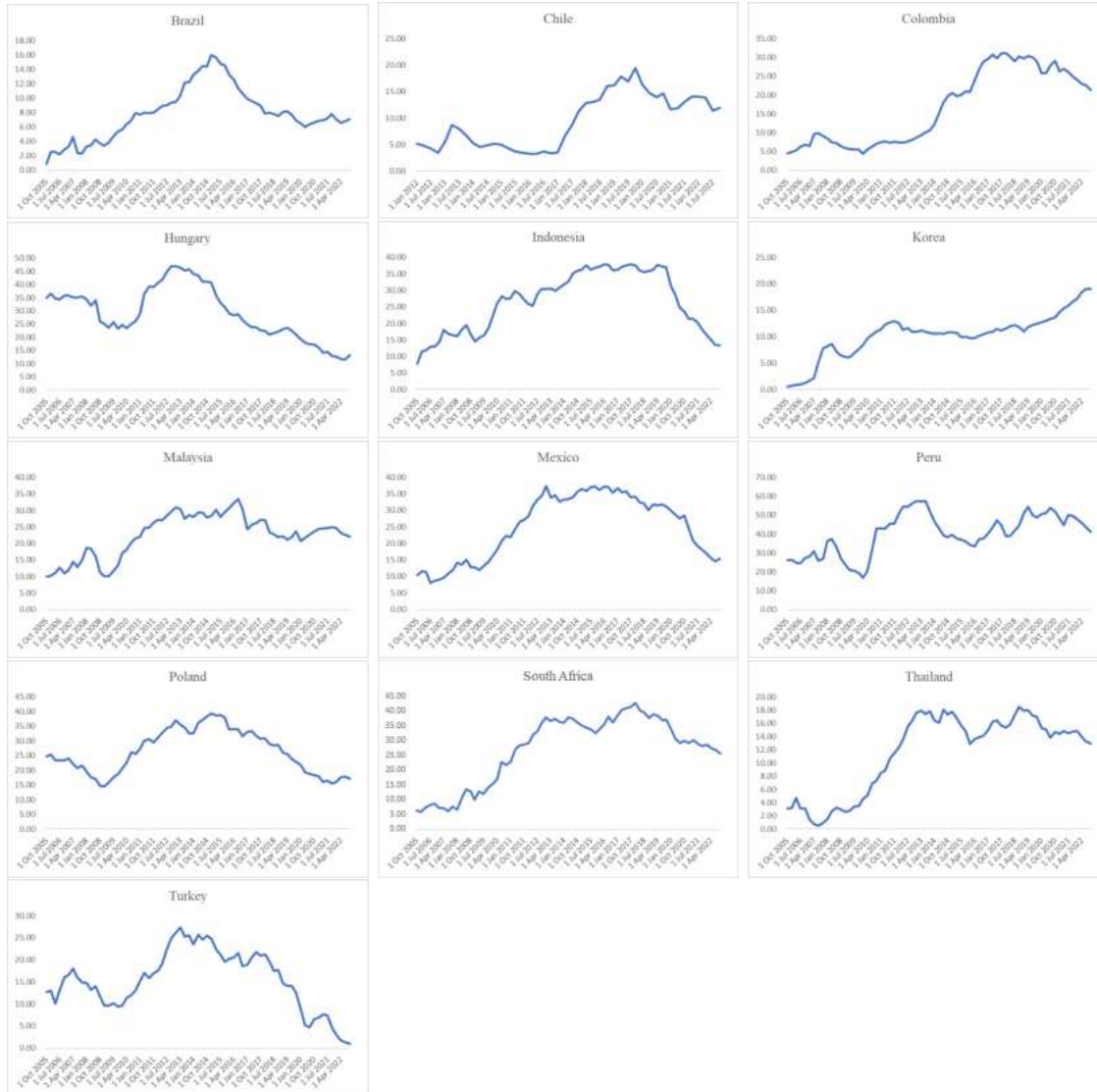
Figure 4: Share of Outstanding Long-Term Sov. Bonds Denominated in Local Currency



banks, and various reports by the BIS. These are all available upon request to the authors.

Brazil. Government bonds are the centerpiece of the securities markets, and comprise the bulk of liquid fixed income instruments. Brazilian investors have a strong preference for high quality short term liquid investments. Brazil's history of economic instability drives investors towards short term liquid investments of the highest credit quality such as overnight repos and short-term government bonds. Most participants in the government bond market are buy-and-hold investors. Investors use interest rate and FX derivatives markets to price and manage risks. These markets are sophisticated, deep, liquid and have sound infrastructure such Central Counterparties (CCPs) to clear transactions. Interest rate derivatives

Figure 5: Share of Local-Currency Long-Term Sov. Bonds Held By Foreign Investors



markets based on both the CDI and SELIC rate exist but almost all trading occurs versus the overnight CDI rate. Bid-offer spreads are low and market depth is high relative to trading government securities hence investors manage risk using derivatives as opposed to trading in cash money market or fixed income instruments. Most money market instruments have shallow markets. The overnight unsecured interbank CDI rate captures few overnight interbank transactions.

Chile. The OTC derivatives market in Chile continues to grow and currently it is estimated that notional amounts of non-delivery forwards (NDFs) on foreign currency and interest rate swaps have reached approximately US\$300 billion, nearly 70% of which is rep-

resented by interest rate swaps. Main actors in the OTC derivatives market are commercial banks, which concentrate approximately 95% of the total notional amount. In July 2015, ComDer, Contraparte Central S.A (hereinafter “ComDer”) was established as a central counterparty for over-the-counter derivatives. ComDer was established as a response of the banking system to the exponential growth of the over-the-counter (OTC) derivatives market and to achieve compliance with international standards and G20 expectations. In particular, ComDer uses good and conservative practices with regard to collateral, e.g. it accepts only cash and debt securities issued by the Central Bank or the National Treasury as collateral, marks collateral and participant positions to market daily, and applies conservative haircuts that also incorporate crisis scenarios thus reducing the need for pro-cyclical adjustments. The interbank money market is unsecured, equivalent to less than three percent of bank assets and under one percent of liabilities. Interbank trading takes the unusual form of the issuance of CDs (42 percent, down from 59 percent in 2018), unsecured deposits (42 percent, up from 21 percent in 2018), and bank bonds (16 percent, down from 20 percent in 2018).

Colombia. Colombia’s capital markets reflect mainly activity in government debt and equity markets, with capitalization reaching 60 percent of GDP at end-2011. Non-government fixed income remains undeveloped (4 percent of GDP) and dominated by financial sector issues. The derivatives clearing house CRCC has prospered since 2008 without access to intra-day central bank credit, but it is growing very fast and becoming more systemically important. As a central counterparty, CRCC interposes itself in the accepted operations becoming buyer of every seller and seller of every buyer, thus mitigating the counterparty risk and generating operational efficiencies to the market through reliable risk management systems. CRCC provides us with an appropriate risk management in order to guarantee security and efficiency in the Colombian capital market.

Mexico. The government bond market is highly liquid and is also the most liquid of Mexico’s securities markets. The liquidity of government securities is underpinned by a diverse investor base and the availability of interest rate derivatives for risk management. In contrast to the deep and liquid public debt market, corporate debt and equity markets are characterized by few issuers. Institutional investors, such as pension funds and mutual funds, hold most of their assets in fixed-income securities, mostly government bonds. The

most important derivatives market in Mexico is the over-the-counter (OTC) derivatives market. Foreign exchange derivatives remain the most important segment of the OTC derivatives market, followed by interest rate derivatives. The TIE swap is the most important contract in the OTC interest rate derivatives market. Mexico's onshore and offshore OTC derivatives markets are fully integrated. As a result, participants in these markets can benefit from lower costs and greater liquidity and price transparency relative to the local exchange-traded market. Transactions in the OTC market can be conducted either with a domestic or foreign bank in Mexico, or directly with an offshore dealer. Foreign investors prefer trading offshore since they feel more comfortable with documentation and the credit quality of offshore counterparties. Counterparty risk to end users in these markets is relatively low since the major derivatives dealers are investment-grade institutions. Among domestic corporates and investors, access to the OTC derivatives market is generally limited to those with high credit ratings and conducting relatively large transactions.

South Africa. The South African financial markets are the most developed and liquid in Africa and well developed by global standards, as well, reflecting credible and independent policy making, a diverse economy and strong financial institutions. The government bond and interest rate swap yield curves go out to 20 years. The central bank operates independently and at a high capacity, providing a sound footing for market functioning. The secondary market for Treasury Bills (short-term) is underdeveloped compared to its size and importance. Although the outstanding amount is high, around 450 billion rand, there are no market making arrangements in place—despite the presence of a primary dealer system in South Africa. The derivatives markets are dominated by interest rate derivatives, mainly forward rate agreements and interest rate swaps. These are dominated by foreign investors and global banks. ISDA Market Agreements with Credit Support Annex (CSA) and Global Repo Market Agreements (GMRA) are market standards among major market participants, but a significant portion of transactions between local banks and their domestic counterparts are underdocumented. However, currently there is no central clearing counterparty for OTC derivative trades. The interbank market is primarily unsecured which increases credit risk in transactions and tend to close banks with less strong credit standing out of the market.

Turkey. Capital markets and long-term savings are underdeveloped, while foreign in-

vestors have withdrawn from Turkey. At end-2021, outstanding public and private debt amounted to 40 and 7½ percent of GDP respectively, while stock market capitalization was 30 percent of GDP, below peers. Government bonds account for more than 90 percent of debt securities. SOBs own around 38 percent of government bonds, while the CBRT holds just over 4 percent. Foreign investor holdings of public debt declined from 20 percent in 2017 to just 1.6 percent in mid-2022. Counterparty concern appears less of an issue for Turkish banks than underlying swap rollover or interest rate risks. London-based global banks are the main swap counterparts, with 89 percent of the total swap transactions. While swap contracts are done on an OTC basis, counterparts typically have credit support annex (CSA) agreements and daily margining between each other to mitigate credit risks. Measures to support the development of derivatives markets and improve issuance regulations, disclosure, and governance standards for capital market products would be helpful. Policies to support the consolidation of an interest rate derivatives market and stimulate liquidity in government and corporate bond markets could facilitate hedging against interest rate fluctuations and enhance demand for capital market products.

Poland. The government bond market has steadily expanded while the corporate bond market has remained shallow. As the size of the government bond market increased from PLN533 billion in 2013 to PLN621 billion in 2017, market infrastructure improved and the interest of domestic and foreign investors was sustained. Foreign investors are drawn to Poland for its relatively deep and liquid government securities market, which enables them to take positions not just on Poland but indirectly on the region as well. The risk of sudden reversal is mitigated by a relatively stable and diversified bond investor base, high domestic liquidity, and sound hedging practices. The interest rate swaps are relatively common in Poland. According to market participants, IRS transactions between banks as well as banks and large players usually use CCP to mitigate counterparty credit risk (the majority of PLN-denominated interest rate derivatives have to be centrally cleared due to the clearing obligation introduced by the EU law), but this is not yet the case with respect to transactions between banks and non-financial medium and small entities, where both parts can be exposed to counterparty credit risk. It is worth mentioning that the scope of the clearing obligations is gradually widened. At the moment, it applies to

large financial institutions (encompassing virtually all interbank transactions and the vast majority of turnover in general) and starting from June 2019, it will also apply to Category 3 entities – i.e. remaining, smaller, financial counterparties. Moreover, notwithstanding the aforementioned clearing obligation credit risk in bilaterally cleared transactions is mitigated by the mandatory exchange of collateral (in accordance with EU law provisions and ISDA recommendations). Currently all counterparties are obliged to provide variation margins covering risk associated with valuation of their derivatives transactions. Larger entities are additionally required to exchange initial margins (the scope of the initial margin requirement is also gradually increased to encompass all entities falling under EMIR).

Thailand. The interest rate swap market in Thailand is liquid and widely utilised by local and foreign participants to manage their interest rate exposures both from investment and borrowing. It also allows players to take an interest rate view without much need for cash funding, and hence represents a good way for nonresidents to take rate exposures in Thailand. Market contacts report that the IRS market is very active. Market participants also note that the nondeliverable IRS has become very active in the last 6–12 months, partly as a consequence of the restrictions placed on foreign investors’ entry into the Thai market directly. Foreign banks with a presence in Thailand appear to intermediate effectively between both the on- and offshore markets, equating prices across both. Thai domestic banks have a limited presence in the international swaps market, as they typically do not meet international counterparty risk standards. Although swap market is an OTC market it has developed in many aspects. For instance, reducing counterparty risk, increasing transparency and improving infrastructure. In particular, when the International of Swaps and Derivatives (ISDA) was established for improving the swap market to be more standard.

Philippines. Onshore interest rate swaps were launched on 18 August 2003 among six foreign banks (Deutsche Bank, HSBC, Standard Chartered, Citibank, JP Morgan and ING). The floating leg is the three-month PHIREF or Philippine Interbank Reference Rate. The PHIREF is the implied peso rate derived from all dealt USD/PHP swap and forward interbank transactions. There is not much information on central clearing, ISDA agreements, etc.

Malaysia. Most banks hold some government securities for collateral and liquidity

reasons. OTC derivatives are largely plain vanilla. In 2011, foreign exchange derivatives comprised 70 percent of total OTC derivatives, followed by interest rate derivatives which account for 27 percent. Trades on both the securities and derivatives exchanges are cleared and settled through central clearing houses that have detailed and transparent provisions designed to protect the markets against a default by any participant. The widely accepted market standard is the International Swap & Derivatives Association, Inc. (ISDA). There is one securities exchange (Bursa Malaysia Securities) and one derivatives exchange (Bursa Malaysia Derivatives) operating in Malaysia under authorization granted by the Minister. Both exchanges, along with the securities and derivatives clearing houses, central securities depository and the operator of an electronic trading platform for the trading and reporting of bonds are subsidiaries of the Bursa Malaysia Berhad, a holding company also authorized by the Minister under the CMSA. To improve accessibility, Bursa Malaysia Derivatives entered into a strategic partnership with Chicago Mercantile Exchange (CME) in 2010. This arrangement includes the licensing of CPO futures settlement prices and global distribution of Bursa Malaysia Derivatives' products through the CME's Globex electronic trading platform.

Hungary. The Hungarian government securities market is now a developed and mature market. Hungarian government bonds are issued for five benchmark maturities, by the Government Debt Management Agency. The majority of (fixed-for-floating) forint interest rate swap transactions are concluded by foreign banks (typically headquartered in London) with one another and with their similarly non-resident clients, e.g. various fund managers. Interest rate swap transactions by themselves contain a minimal non-performance risk. The default risk is further reduced by the netting of interest payments, which is even further reduced by the application of daily margining, which has become widespread in the forint interest rate swap market. According to market participants' estimates, the turnover of the forint interest rate swap market in London is approximately 2-4 times higher than that of the domestic market. Anecdotal information suggests that all significant investment banks in London participate in the London-based forint interest rate swap market, although only some of them can be considered as real market makers. Market makers in the domestic market include some of the major commercial banks. They mainly trade with non-resident

clients, mostly banks, and only a relatively small portion of their turnover is with domestic partners. Moreover, in most cases the domestic partners are also banks. Apart from them only a certain number of fund managers and insurance corporations are present in the market, while other companies practically do not conclude interest rate swaps in Hungary.

Appendix C Data Sources

Recall from the main text the expression for the dollar convenience yield:

$$\lambda_t^{T,f} - \lambda_t^{US,f} = y_t^{US} - (y_t^T - \rho_t) + (l_t^T - l_t^{US}) + \Phi_t^{FC} - \tau_t + (\xi_t^{US,f} - \xi_t^{FC,f}) \quad (12)$$

The sources for each component are the following:

Bond yields and forward premia. We used data from the Resolution Funding Corporation (Refcorp) bonds for various maturities for yields of non-Treasury-safe dollar bonds. As suggested by Longstaff (2004), Refcorp bonds are effectively guaranteed by the U.S. government and are subject to the same taxation, but are not as liquid as Treasuries. As in Longstaff (2004), we measured the yields by taking the differences between the constant maturity on the Bloomberg Fair Value curves for Refcorp zero-coupon bonds. Maturities available are 6-month, 1-, 2-, 3-, 4-, 5-, 7-, 10-, and 20-year. For robustness, we also used the yields for Aaa corporate bonds, which Krishnamurthy and Vissing-Jorgensen (2012) argued have very low default rates but are not as liquid as Treasuries. Data on these corporate bond spreads are available in FRED but only provide a 20-year maturity benchmark.

The other two yields for non-Treasury safe dollar bonds correspond to the ICE Bank of America AAA and BBB US Corporate Index. These track the performance of US dollar-denominated corporate debt issued in the US domestic market, with AAA and BBB credit ratings, respectively. They include all maturities greater than one year. The series were retrieved from FRED, Federal Reserve Bank of St. Louis.

The value of the forward premium for each country was taken from the database of Du et al. (2018). The authors provide estimations of CIP deviations of sovereign bonds for ten developed and 18 developing countries to U.S. Treasuries. The data are at a daily

frequency between approximately 2000 and March 9, 2021, although the start date varied among countries. Data are available for maturities at 3-months, 1-, 2-, 3-, 5- and 10-years.

We focused on their observations of developing countries. Their bond yields data came from Bloomberg and Thomson Reuters. Since forward contracts are, in general, not very liquid, they computed $\rho_{i,t}$ from a hedging strategy involving interest rate swaps and cross-currency swaps, according to the formula $\rho_{i,n,t} = irs_{i,n,t} + bs_{i,n,t} - irs_{US,n,t}$. $irs_{i,n,t}$ is the n -year interest rate swap for exchanging fixed currency i cash flows into the floating interbank rate benchmark in country i . $bs_{i,n,t}$ is the n -year cross-currency basis swap rate for exchanging the floating benchmark interbank rate in country i for the U.S. Libor rate, and $irs_{US,n,t}$ is the n -year U.S. Libor swap rate for exchanging fixed dollar cash flows into the U.S. Libor rate. The combination of these three swaps eliminates all floating cash flows. At the inception and maturity of the swap, only fixed cash flows remain between local currency and U.S. dollars, which exactly replicates an n -term forward contract.

Default risk differentials ($l_t^{US} - l_t^T$). We proxied $l_{i,t}$ with data on CDS spreads. We obtained the CDS spread series for EMEs' sovereign bonds of different maturities from Bloomberg at a daily frequency. However, some caveats apply. First, we used the CDS spreads for foreign-currency debt, as their data are more widely available and show greater liquidity than local-currency CDS. Therefore, we assumed that the risk of default on foreign-currency debt also applies to local-currency bonds. As discussed in Du and Schreger (2016), this assumption is not much different from reality as default events in EMEs since the late 1990s show that the incidence of default on domestic-currency debt is comparable with the incidence of external foreign-currency defaults.

Spread between swapped local-currency bond and foreign-currency bond (Φ_t^{FC}). We used the data from Du and Schreger (2016). For y_t^{FC} , we used the Bloomberg Fair Value curves (BFV) for the prices of foreign-currency sovereign bonds for each EME. These are at par yield curves, so they must be adjusted to represent zero-coupon yields. BFV prices are not available for some of the countries. In those cases, we estimated prices by collecting data for each bond and computing the overall zero-coupon yield curve using the methodology of Nelson and Siegel (1987).

Liquidity risk of forwards contracts (τ_t). τ_t is measured as half the bid-ask spread

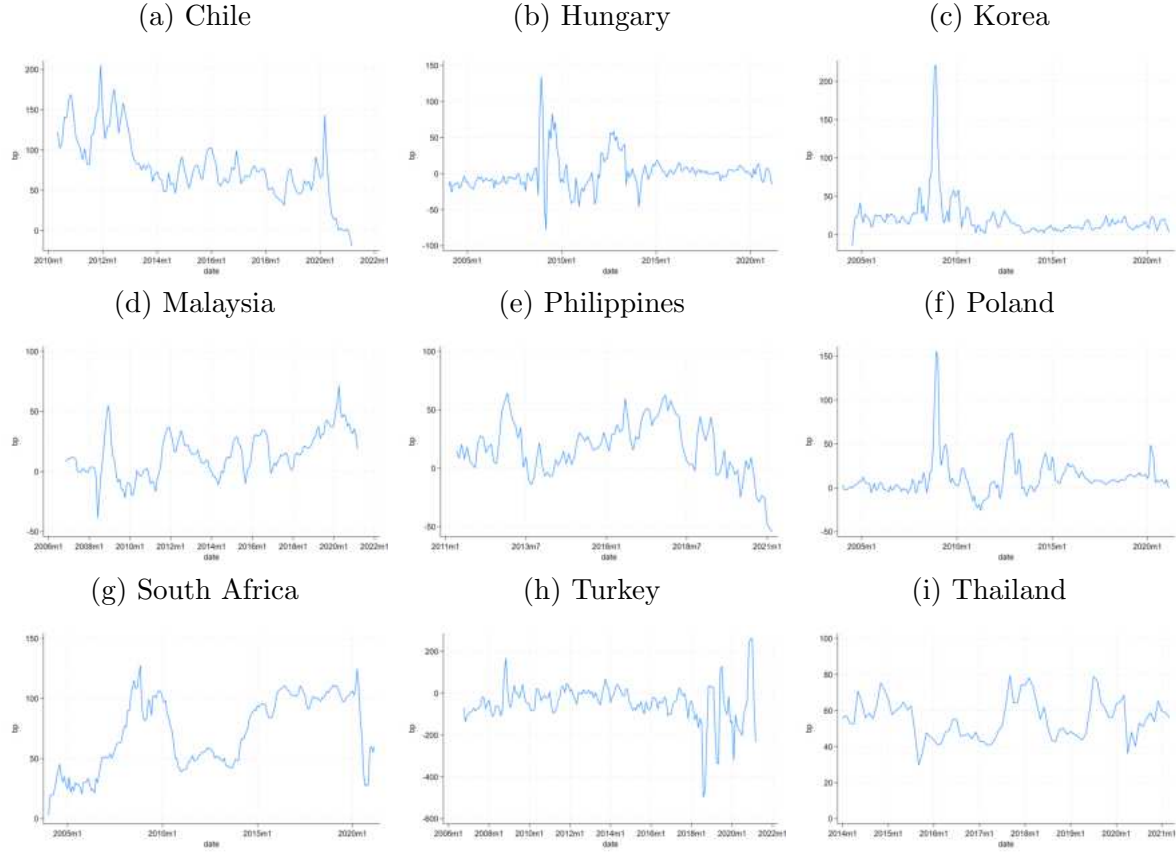
of each cross-currency swap. Daily data is available on Bloomberg. For Brazil, Colombia, Indonesia, Peru, Philippines, and Turkey, the cross-currency swap is the non-deliverable swap between the fixed local rate and the floating U.S. Libor. For Chile, Mexico, and South Africa, it corresponds to the bid-ask spread of the interest rate swap used to construct the cross-currency swap in Du and Schreger (2016).

Domestic convenience yield The yield on the 1-year local-currency sovereign bond comes from the dataset in Du et al. (2018). The private local-currency domestic assets used for each country are listed in Table 6. All yields are for the 1-year maturity except for Mexico, where only the 9-month maturity was available.

Table 6: Private local-currency domestic assets (Bloomberg tickers)

Country	Interbank rate	Interest rate swap
Brazil	–	PREDI 360 Index
Chile	CLTN360N Index	CHSWP1
Colombia	–	CLSWIB1 Curncy
Hungary	BUBOR01Y Index	HFSW1
Indonesia	–	–
Korea	KRBO12M Index	KWSWO1 Curncy
Malaysia	KLIB12M Index	MRSWNI1 Curncy
Mexico	–	MPSW1A Curncy
Peru	–	–
Philippines	PREF1YR Index	PPSWO1 Curncy
Poland	WIBR1Y Index	PZSW1 Curncy or PZOSW1
Thailand	BOFX1Y Index	TBSWNI1 or TBSWI1
South Africa	JIBA12M Index	SASW1 Curncy
Turkey	TRLXB1Y Index	TYSWF Curncy

Figure 6: Safety Premium by Country (1-year Maturity)



Notes: Figure shows each country's safety premium.

Appendix D Safety and Liquidity Premium by Country

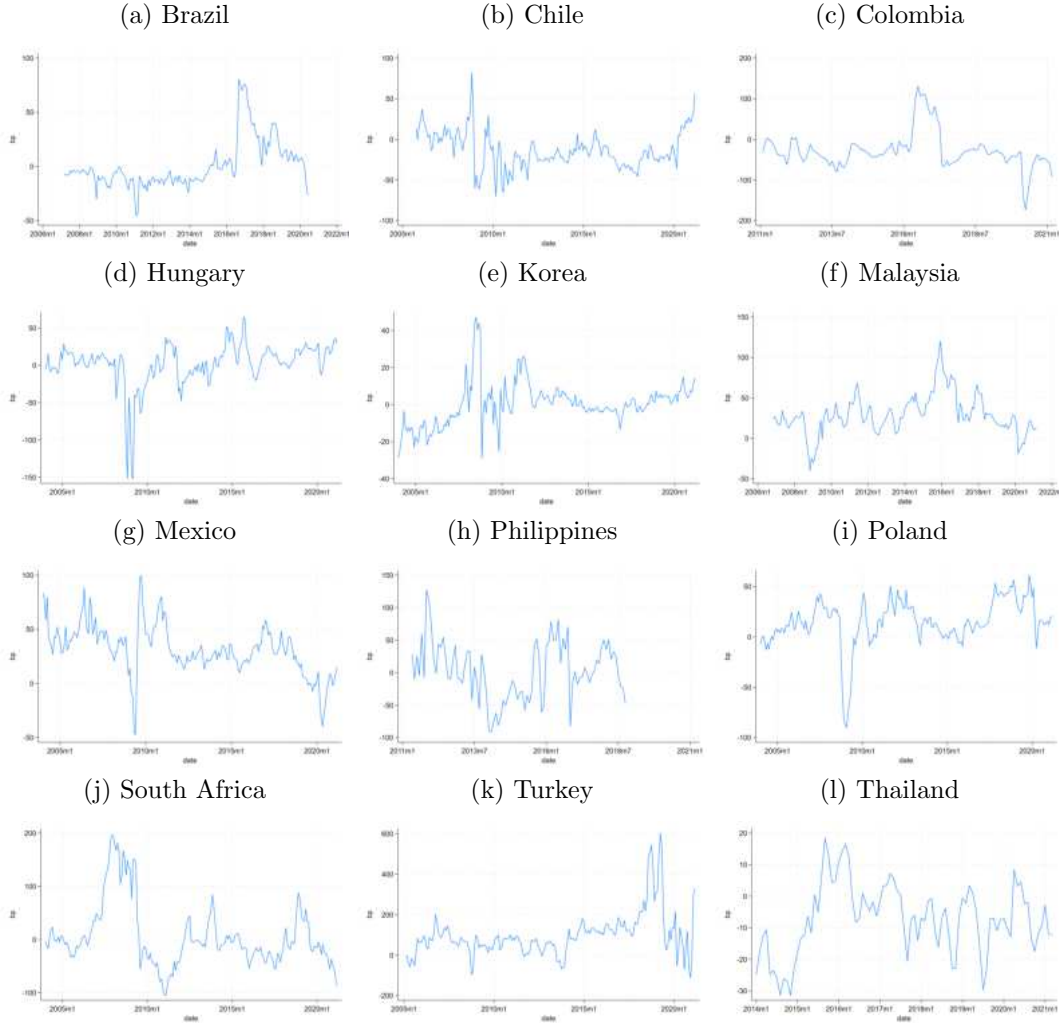
Figure 6 shows the safety premium of Section 3.1 by country.

Figure 7 shows the liquidity premium of Section 3.1 by country.

Appendix E A Model of Convenience Yields and Safe-Haven Status

This model shows that, taking as given the safe-haven status of an asset, its convenience yield should positively correlate with the corresponding source of risk for which it is a safe-haven. Therefore, it is not about the endogenous choice among different safe assets. The

Figure 7: Liquidity Premium by Country (1-year Maturity)



Notes: Figure shows each country's liquidity premium.

model is set up in a way that the safe-haven status derives from the liquidity value in times of distress. It is simple to extend the framework to allow for the safe-haven to provide safety services, and this latter case is discussed at the end of this Section.

Time is discrete and runs for three periods: $t = 0, 1, 2$. An investment decision is made under uncertainty at $t = 0$, shocks are realized in $t = 1$ and payoffs materialize in the last period $t = 2$. Investors and intermediaries allocate capital across issuers of identical risk-free bonds. The government and firms each issue a fixed number of sovereign and corporate bonds, respectively, that pay \$1 with certainty in $t = 2$ and the model can be interpreted equivalently as nominal or real. The model endogenously determines the price of these two types of bonds in $t = 0$ and $t = 1$.

There is a unit measure of investors, a fraction $\lambda_t \in (0, 1)$ of whom become liquidity-constrained each period and must liquidate their bond holdings. The remainder $1 - \lambda_t$ are unconstrained and act as secondary-market buyers.

There are two one-period zero-coupon assets. A sovereign bond B_t^s , which is liquid and accepted broadly in resale markets, and a corporate bond B_t^c , which is harder to trade or pledge. Each bond pays 1 unit of consumption next period (both have the same credit risk). Prices at time t are P_t^s, P_t^c .

E.1 Stochastic Risk Structure

There are two sources of time-varying uncertainty. First, a global shock: $\varepsilon_t^G \sim N(0, \sigma_{G,t}^2)$; and a local shock for country i : $\varepsilon_{i,t}^L \sim N(0, \sigma_{L,it}^2)$.

We assume the volatility of each shock evolves according to:

$$\begin{aligned}\sigma_{G,t+1}^2 &= \rho_G \sigma_{G,t}^2 + (1 - \rho_G) \bar{\sigma}_G^2 + \nu_t^G \\ \sigma_{L,i,t+1}^2 &= \rho_L \sigma_{L,it}^2 + (1 - \rho_L) \bar{\sigma}_L^2 + \nu_{i,t}^L\end{aligned}$$

where $\nu_t^G, \nu_{i,t}^L$ are white noise innovations. These stochastic volatilities are observed by all agents and serve as state variables.

To give some interpretation, $\sigma_{G,t}^2$ captures global uncertainty (e.g., from monetary, financial, or geopolitical shocks) while $\sigma_{L,it}^2$ reflects country-specific economic or political risk.

Brunnermeier and Pedersen (2009) show that dealers' margin requirements are procyclical (higher when volatility rises), which tightens their funding constraint and reduces their intermediary capacity. We summarize this fact in a reduced-form way by assuming the following:

- The fraction of agents forced to sell is a function of volatility: $\lambda_t = \lambda(\sigma_{G,t}^2, \sigma_{L,it}^2)$, with $\partial \lambda / \partial \sigma^2 > 0$
- Same for intermediary balance sheet pressure: $\eta_t = \eta(\sigma_{G,t}^2, \sigma_{L,it}^2)$, also increasing in risk.

E.2 Investor Preferences and Portfolio Choice

Each investor maximizes:

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(C_t) \right]$$

subject to:

$$C_t + P_t^s B_t^s + P_t^c B_t^c = I_t$$

where I_t is total income from maturing assets and exogenous sources. Investors internalize the probability λ_t of needing to liquidate at a loss.

E.3 Secondary Market and Intermediaries

At $t = 1$, unconstrained agents act as intermediaries, absorbing distressed sales. They face a balance sheet constraint:

$$\theta^s q_t^s Q_t^s + \theta^c q_t^c Q_t^c \leq K$$

with haircuts $\theta^c > \theta^s$, so sovereign bonds are more easily absorbed.

Market clearing requires:

$$\lambda_t B_t^j = Q_t^j$$

Intermediaries' relative willingness to buy sovereign vs. corporate bonds is not driven by the investors' liquidity shock per se. It's a separate friction. θ^j is the haircut or capital charge per unit of bond j . It needn't come from the same source as investors' liquidity risk. Instead, it could reflect regulatory risk weights (sovereigns often carry low risk weight), repo eligibility and margin requirements, or market-making capacity (dealers hold more sovereigns because they're easier to hedge).

E.4 Equilibrium Pricing

Intermediaries price assets in secondary markets ($t = 1$):

$$q_t^j = \mathbb{E}_t[M_{t+1}] - \eta_t \theta^j$$

where η_t is the Lagrange multiplier on the balance sheet constraint. Sovereign bonds get a smaller discount (since $\theta^c > \theta^s$), so $q_t^s > q_t^c$ when $\eta_t > 0$. This captures the fire-sale value in the secondary market.

Investors buy bonds at time $t = 0$, knowing that, with probability $1 - \lambda_t$, they hold it to maturity and get $\mathbb{E}_t[M_{t+1}]$, and with probability λ_t , they are forced to sell at the secondary market price q_t^j .

The primary market prices ($t = 0$) are:

$$P_t^j = (1 - \lambda_t)\mathbb{E}_t[M_{t+1}] + \lambda_t q_t^j = \mathbb{E}_t[M_{t+1}] - \lambda_t \eta_t \theta^j$$

Investors care about the expected resale price of whatever they buy. They don't need to know why intermediaries pay more for sovereigns—just that they do. That's embedded in the parameter θ^j . Investors face a risk of forced sales, and so value collateral that will command better fire-sale prices. Intermediaries absorb those sales but only up to the point their capital or regulatory buffers allow; they demand a discount proportional to θ^j .

Now, the differential convenience yield can be obtained as the spread between the two bonds: $DCY_t = y_t^c - y_t^s$, where yields are $y_t^j = -\log P_t^j$. So we translate:

$$DCY_{it} = y_t^c - y_t^s = \log P_0^s - \log P_0^c$$

$$\text{given } P_t^j = \mathbb{E}_t[M_{t+1}] - \lambda_t \eta_t \theta^j.$$

In this model, safe-haven status means that, when risk increases, investors want to hold the asset more, because they are aware that intermediaries are more willing to buy these assets over others. This means that its price increases or drops less than others in crises. So, safe-haven assets are those whose marginal utility of holding the asset increases in bad times. Sovereigns are “safe havens” for investors because they know that under stress, intermediaries will stand ready to buy them at relatively good prices.

The convenience yield is the premium investors are willing to pay (accepting a lower yield) for an asset that provides higher liquidity. Liquidity refers to the ability of the asset to be sold quickly, with minimal price impact, when investors need to rebalance. In this framework, liquidity is especially valuable when there are fire sales or when intermediary

balance sheets are tight. In such environments, more liquid assets command higher prices (via higher convenience yields).

Intermediaries' willingness to buy sovereigns is an independent friction. Investors simply rationally prefer the asset that will trade cleanly in a fire sale, and that preference translates into a convenience yield in normal times. In recent papers studying convenience yields, they also arise from the interaction of two frictions. It is common to combine liquidity shock with search frictions (Coppola, Krishnamurthy, and Xu, 2024). We prefer to rely on a balance sheet constraint to keep the model simple and highlight the key mechanisms.

E.5 From the Model to the Empirical Regression

Take the expression for the differential convenience yield:

$$DCY_t = y_t^c - y_t^s = \log P_0^s - \log P_0^c \approx \frac{\lambda_t \eta_t (\theta^c - \theta^s)}{\mathbb{E}_t[M_{t+1}]} \quad (13)$$

Call the constant $\kappa \equiv \frac{(\theta^c - \theta^s)}{\mathbb{E}_t[M_{t+1}]}$.

Then,

$$DCY_t = \kappa \lambda_t \eta_t \quad (14)$$

As explained above, the frequency of distressed sales (λ_t) and the severity of intermediary strain (η_t) are increasing functions of volatilities. A first-order (linear) approximation is

$$\lambda_t \eta_t \approx \underbrace{a}_{\lambda\eta \text{ at baseline}} + b_1 \sigma_{G,t}^2 + b_2 \sigma_{L,i,t}^2 \quad (15)$$

Substitute back into the theoretical DCY_t formula:

$$DCY_t = \kappa [a + b_1 \sigma_{G,t}^2 + b_2 \sigma_{L,i,t}^2] + \varepsilon_{it} \quad (16)$$

where ε_{it} captures higher-order terms (omitted nonlinearities) and residual noise.

Re-labeling the constants and collecting terms gives the following empirical regression:

$$\underbrace{DCY_t}_{\text{empirical yield spread}} = \underbrace{\alpha_i}_{\text{countryFE}} + \underbrace{\gamma_i}_{\text{timeFE}} + \beta_1 \sigma_{G,t}^2 + \beta_2 \sigma_{L,i,t}^2 + \varepsilon_{it} \quad (17)$$

where $\beta_1 = \kappa b_1$ and $\beta_2 = \kappa b_2$.

If macro risk raises the frequency of distressed sales and tightens intermediary constraints, then:

$$\frac{\partial DCY_{it}}{\partial \sigma_{G,t}^2} > 0, \quad \frac{\partial DCY_{it}}{\partial \sigma_{L,i,t}^2} > 0$$

In the empirical regression, this links to the following interpretation. If $\beta_1 > 0$, as global volatility rises, intermediaries face a higher fire-sale risk, and investors pay up for liquidity; then the sovereign bond is considered a “safe haven” from global risk. If $\beta_2 > 0$, as local volatility rises, similar logic applies at the country level, and the sovereign bond is a “safe haven” from local turmoil. What about the corporate bond, or when either of the β s are negative? This means the bond does not work as a safe haven. In the model, even though risk increases, investors are aware that intermediaries are unlikely to be willing to buy this bond, so they do not bid up the price.

Notice that this is not a model about the endogenous determination of a safe-haven (as in He, Krishnamurthy, and Milbradt (2019) or Brunnermeier, Merkel, and Sannikov (2024)). Rather, it suggests that, given that a bond is a safe haven, its convenience yield should positively correlate with the corresponding source of systematic risk.

An alternative would be to focus on safety services. Instead of forced sales, investors face the risk that, at maturity, some bonds will offer lower nominal repayment than others. If there is demand for collateral to mitigate counterparty risk in derivatives and settlement systems, the collateral in these transactions must be extremely safe, thereby driving a demand for the safety attribute. In addition, investors might have a special demand for safe nominal repayment if they borrowed at $t = 0$ and face nominal obligations at $t = 2$. More generally, any motive driving a preferred habitat for extremely safe bonds could drive a safety premium..

Internet Appendix

IA. Proofs of Propositions 1-3 (Section 2)

Proof of Proposition 1. This proof is similar to the one in Krishnamurthy and Vissing-Jorgensen (2012). Denote the domestic price level at date t as Q_t . If the investor buys a zero-coupon nominal domestic sovereign bond for a price P_t^T , her real holdings θ_t^T rise by P_t^T/Q_t . The first order condition for this bond holdings is, then,

$$-\frac{P_t^T}{Q_t}u'(C_t) + \beta E_t \left[\frac{P_{t+1}^T}{Q_{t+1}}u'(C_{t+1}) \right] + \frac{P_t^T}{Q_t}v'(\theta_t^d/\text{GDP}_t^d)u'(C_t) = 0 \quad (18)$$

Define the pricing kernel for nominal payoffs as

$$M_{t+1} = \beta \frac{u'(C_{t+1})}{u'(C_t)} \frac{Q_t}{Q_{t+1}} \quad (19)$$

so that, in the absence of default risk, we would have:

$$P_t^T = E_t[M_{t+1}P_{t+1}^T\Lambda_{t+1}^{T,d}] \quad (20)$$

where $\Lambda_t^{T,d} \equiv 1/(1 - v'(\theta_t^d/\text{GDP}_t^d))$ captures the marginal benefits investor d derives from these local-currency sovereign bonds of the EME. A positive marginal value of convenience, $v'(\cdot)$, raises $\Lambda_t^{T,d}$, and therefore raises the price of the bond, P_t^T .

Suppose that the EME sovereign can default next period with probability π_t^T , and L_{t+1}^T measures the amount of losses suffered in default (a random variable). If the bond does not default, it is worth P_{t+1}^T . Then, its price satisfies,

$$P_t^T = \pi_t^T E_t[M_{t+1}\Lambda_{t+1}^{T,d}(1 - L_{t+1}^T)|\text{Default}] + (1 - \pi_t^T)E_t[M_{t+1}P_{t+1}^T\Lambda_{t+1}^{T,d}|\text{No Default}] \quad (21)$$

For simplicity, assume continuously compounded yields and consider the case of one-period bonds (so $P_{t+1}^T = 1$). Define $\tilde{L}_{t+1}^T$ as a random variable that is equal to zero if there is no default and equal to L_{t+1}^T if there is a default. Then, the expression for the price of the

bond is

$$\begin{aligned} e^{-y_t^T} &= P_t^T = E_t[M_{t+1}\Lambda_{t+1}^{T,d}] - E_t[M_{t+1}]E_t[\tilde{L}_{t+1}^T] - \text{cov}_t[M_{t+1}, \tilde{L}_{t+1}^T] - \text{cov}_t[\Lambda_{t+1}^{T,d}, \tilde{L}_{t+1}^T] \\ &\approx e^{\lambda_{t+1}^{T,d} - \pi_t^T(E_t[L_{t+1}^T] + \text{cov}_t[M_{t+1}, \tilde{L}_{t+1}^T]/E_t[M_{t+1}] - \text{cov}_t[\lambda_{t+1}^{T,d}, \tilde{L}_{t+1}^T]/E_t[M_{t+1}])} E_t[M_{t+1}] \end{aligned} \quad (22)$$

where $\lambda_t^{T,d} \approx v'(\theta_t^d/\text{GDP}_t^d)$ and $\text{cov}_t[M_{t+1}, \tilde{L}_{t+1}^T]/E_t[M_{t+1}]$ is a risk premium if default events coincide with bad states. Take logs on both sides to get :

$$y_t^T \approx y_t^{rf} - \lambda_t^{T,d} + l_t^T - \xi_t^{T,d} \quad (23)$$

where $y_t^{rf} = -\log M_{t+1}$ (no arbitrage condition); $l_t^T = \pi_t^T(E_t[L_{t+1}^T] + \text{cov}_t[M_{t+1}, \tilde{L}_{t+1}^T]/E_t[M_{t+1}])$ denotes the compensation for default (expected losses plus premium); $\lambda_t^{T,d}$ is the convenience yield (how much the total yield is reduced because of the marginal services provided by the bond); and $\xi_t^{T,d} = \text{cov}_t[\lambda_{t+1}^{T,d}, \tilde{L}_{t+1}^T]/E_t[M_{t+1}]$ denotes the covariance between default risk and the convenience yield.

The decomposition of the yield of the private asset follows the same logic,

$$y_t^P \approx y_t^{rf} - \lambda_t^{P,d} + l_t^P - \xi_t^{P,d} \quad (24)$$

Take the spread between the two yields of the same maturity to get:

$$y_t^P - y_t^T \approx (\lambda_t^{T,d} - \lambda_t^{P,d}) + (l_t^P - l_t^T) + (\xi_t^{T,d} - \xi_t^{P,d}) \quad (25)$$

□

Proof of Proposition 2. Denote the US price level at date t as $Q_t^\$$. If the investor buys a zero-coupon nominal non-Treasury safe U.S. bond for a dollar price P_t^{US} , her real holdings θ_t^{US} rise by $P_t^{US}/Q_t^\$$. The first order condition for this bond holdings is then

$$-\frac{P_t^{US}}{Q_t^\$} u'(C_t) + \beta E_t \left[\frac{P_{t+1}^{US}}{Q_{t+1}^\$} u'(C_{t+1}) \right] + \frac{P_t^{US}}{Q_t^\$} v'(\theta_t^f/\text{GDP}_t^f) u'(C_t) = 0 \quad (26)$$

Define the pricing kernel for nominal payoffs as

$$M_{t+1} = \beta \frac{u'(C_{t+1})}{u'(C_t)} \frac{Q_t^\$}{Q_{t+1}^\$} \quad (27)$$

so that, in the absence of default risk, we would have:

$$\begin{aligned} P_t^{US} &= E_t[M_{t+1}P_{t+1}^{US}] + P_t^{US}v'(\theta_t^f/\text{GDP}_t^f) \Rightarrow \\ P_t^{US} &= E_t[M_{t+1}P_{t+1}^{US}\Lambda_{t+1}^{US,f}] \end{aligned} \quad (28)$$

where $\Lambda_t^{US,f} \equiv 1/(1 - v'(\theta_t^f/\text{GDP}_t^f))$ captures the marginal benefits investor f derives from these non-Treasury safe bonds. A positive marginal value of convenience, $v'(\cdot)$, raises $\Lambda_t^{US,f}$, and therefore raises the price of the bond, P_t^{US} .

To add default risk, suppose that the issuer may default next period with probability π_t and, in default, pays $1 - L_{t+1}^{US}$, where L_{t+1}^{US} measures the amount of losses suffered in default (and is a random variable). If the bond does not default, it is worth P_{t+1}^{US} . Then, its price satisfies,

$$P_t^{US} = \pi_t E_t[M_{t+1}\Lambda_{t+1}^{US,f}(1 - L_{t+1}^{US})|\text{Default}] + (1 - \pi_t)E_t[M_{t+1}P_{t+1}^{US}\Lambda_{t+1}^{US,f}|\text{No Default}] \quad (29)$$

For simplicity, assume continuously compounded yields and consider the case of one-period bonds (so $P_{t+1}^{US} = 1$). Define $\tilde{L}_{t+1}^{US}$ as a random variable that is equal to zero if there is no default and equal to L_{t+1}^{US} if there is a default. Then, the expression for the price of the bond is

$$\begin{aligned} e^{-y_t^{US}} &= P_t^{US} = E_t[M_{t+1}\Lambda_{t+1}^{US,f}] - E_t[M_{t+1}]E_t[\tilde{L}_{t+1}^{US}] - \text{cov}_t[M_{t+1}, \tilde{L}_{t+1}^{US}] - \text{cov}_t[\Lambda_{t+1}^{US,f}, \tilde{L}_{t+1}^{US}] \\ &\approx e^{\lambda_{t+1}^{US,f} - \pi_t(E_t[L_{t+1}^{US}] + \text{cov}_t[M_{t+1}, \tilde{L}_{t+1}^{US}]/E_t[M_{t+1}]) - \text{cov}_t[\lambda_{t+1}^{US,f}, \tilde{L}_{t+1}^{US}]/E_t[M_{t+1}]} E_t[M_{t+1}] \end{aligned} \quad (30)$$

where $\lambda_t^{US,f} \approx v'(\theta_t^f/\text{GDP}_t^f)$ and $\text{cov}_t[M_{t+1}, \tilde{L}_{t+1}^{US}]/E_t[M_{t+1}]$ is a risk premium if default events coincide with bad states. Take logs on both sides to get :

$$y_t^{US} \approx y_{rf,t}^{US} - \lambda_t^{US,f} + l_t^{US} - \xi_t^{US,f} \quad (31)$$

where $y_{rf,t}^{US} = -\log M_{t+1}$ (no arbitrage condition); $l_t^{US} = \pi_t(E_t[L_{t+1}^{US}] + \text{cov}_t[M_{t+1}, \tilde{L}_{t+1}^{US}]/E_t[M_{t+1}])$ denotes the compensation for default (expected losses plus premium); $\lambda_t^{US,f}$ is the convenience yield (how much the total yield is reduced because of the marginal services provided by the bond); and $\xi_t^{US,f} = \text{cov}_t[\lambda_t^{US,f}, \tilde{L}_{t+1}^{US}]/E_t[M_{t+1}]$ denotes the covariance between default risk and the convenience yield. $\square$

Again, denote the price level at date t as $Q_t^\$$. Let the price of the EME sovereign bond be P_t^T . If the investor purchases one unit, her real holdings θ_t^T rise by $P_t^T/Q_t^\$ \times 1/S_t$, where S_t is the nominal exchange rate. The first order condition for holdings of the synthetic bond is

$$-\frac{P_t^T}{Q_t^\$} \frac{1}{S_t} u'(C_t) + \beta E_t \left[\frac{P_{t+1}^T}{Q_{t+1}^\$} \frac{1}{F_{t+1}} u'(C_{t+1}) \right] + \frac{P_t^T}{Q_t^\$} \frac{1}{S_t} v'(\theta_t^f/\text{GDP}_t^f) u'(C_t) = 0 \quad (32)$$

As before, for simplicity, assume one-period bonds, so $P_{t+1}^T = 1$ and the forward rate is a one-period ahead rate, $F_{t+1} = F_t^1$. In the absence of other risks, we would have:

$$\begin{aligned} P_t^T \frac{F_t^1}{S_t} &= E_t[M_{t+1}] + P_t^T \frac{F_t^1}{S_t} v'(\theta_t^f/\text{GDP}_t^f) \Rightarrow \\ P_t^T \frac{F_t^1}{S_t} &= E_t[M_{t+1} \Lambda_{t+1}^{T,f}] \end{aligned} \quad (33)$$

where $\Lambda_{t+1}^{T,f} \equiv 1/(1 - v'(\theta_t^f/\text{GDP}_t^f))$ captures the marginal benefits investor f derives from the bond issued by the EME sovereign.

Recall that the EME sovereign can default next period with probability π_t^T , and L_{t+1}^T measures the amount of losses suffered in default (a random variable). The synthetic bond faces an additional loss upon default. If the sovereign defaults, the currency hedging becomes imperfect, and the investor f loses L_{t+1}^T and still needs to unwind the swap position with unmatched local EME currency cash flows. Regarding positively correlated default and currency risk, the local currency depreciates more upon default than the non-default state. The investor f holding the synthetic bond has a net long position in dollars in the event of default, corresponding to additional currency gains. As a consequence, in the default state, the bond pays $[1 - L_{t+1}^T + L_{t+1}^T(1 - F_{t+1}/S_{t+1})]$.

Du and Schreger (2016) show that the pricing impact of the foreign exchange hedging error, $L_{t+1}^T(1 - F_{t+1}/S_{t+1})$, is precisely equal to $\frac{\text{cov}_t(1 - L_{t+1}^T, 1/S_{t+1})}{E_t(1 - L_{t+1}^T)E_t(1/S_{t+1})}$. We will denote this term

q_t^T and refer to it as the covariance between default and currency risks.

Analogously, assume that the EME sovereign can enact regulations on local-currency assets with probability $\tilde{\pi}_t^T$ (for example, capital controls or currency convertibility restrictions), and this event imposes a loss of K_{t+1} on the investor (a random variable). This loss will also produce a hedging error in the swap position of the investor, as in the case of default losses. Equivalently, define the bond payoff in the event of capital controls as $[1 - K_{t+1} + K_{t+1}(1 - F_{t+1}/S_{t+1})]$. The hedging error term will be exactly equal to $\frac{\text{cov}_t(1-K_{t+1}/S_{t+1})}{E_t(1-K_{t+1})E_t(1/S_{t+1})}$, term which we will denote as p_t and refer to it as the covariance between capital control risk and currency risk.

In the end, the losses in the event of default and regulations impositions are $L_{t+1}^T - q_t^T$ and $K_{t+1} - p_t$, respectively. Define $\tilde{L}_{t+1}^T$ as a random variable that is equal to zero if there is no default and equal to $L_{t+1}^T - q_t^T$ if there is a default. Equivalently, define $\tilde{K}_{t+1}$ as a random variable that is equal to zero if capital controls are not imposed and equal to $K_{t+1} - p_t$ if they are set. Then, the expression for the price of the synthetic bond is

$$\begin{aligned}
e^{-y_t^T + \rho_t} = P_t^T \frac{F_{t+1}}{S_t} &= E_t[M_{t+1}\Lambda_{t+1}^{T,f}] - E_t[M_{t+1}]E_t[\tilde{L}_{t+1}^T] - E_t[M_{t+1}]E_t[\tilde{K}_{t+1}] \\
&\quad - \text{cov}_t[M_{t+1}, \tilde{L}_{t+1}^T] - \text{cov}_t[M_{t+1}, \tilde{K}_{t+1}] - \text{cov}_t[\Lambda_{t+1}^{T,f}, \tilde{L}_{t+1}^T] - \text{cov}_t[\Lambda_{t+1}^{T,f}, \tilde{K}_{t+1}] \\
&\approx e^{\lambda_{t+1}^{T,f} - \pi_t^T (E_t[L_{t+1}^T] + \text{cov}_t[M_{t+1}, \tilde{L}_{t+1}^T]/E_t[M_{t+1}]) + q_t^T - \tilde{\pi}_t^T (E_t[K_{t+1}] + \text{cov}_t[M_{t+1}, \tilde{K}_{t+1}]/E_t[M_{t+1}])} \\
&\quad \times e^{p_t - \text{cov}_t[\Lambda_{t+1}^{T,f}, \tilde{L}_{t+1}^T]/E_t[M_{t+1}] - \text{cov}_t[\Lambda_{t+1}^{T,f}, \tilde{K}_{t+1}]/E_t[M_{t+1}]} \times E_t[M_{t+1}]
\end{aligned} \tag{34}$$

Taking logs on both sides gives:

$$y_t^T - \rho_t \approx y_{rf,t}^{US} - \lambda_t^{T,f} + (l_t^T - q_t^T) + (k_t - p_t) - \xi_t^{T,f} - \psi_t^{T,f} \tag{35}$$

where $y_{rf,t}^{US} = -\log M_{t+1}$; $\lambda_t^{T,f} \approx v'(\theta_t^f/\text{GDP}_t^f)$ is the convenience yield on the local-currency bond; $l_t^T = \pi_t^T (E_t[L_{t+1}^T] + \text{cov}_t[M_{t+1}, \tilde{L}_{t+1}^T]/E_t[M_{t+1}])$ and $k_t = \tilde{\pi}_t^T (E_t[K_{t+1}] + \text{cov}_t[M_{t+1}, \tilde{K}_{t+1}]/E_t[M_{t+1}])$ are the extra yield demanded for default and regulatory losses; and $\xi_t^{T,f} = \text{cov}_t[\Lambda_{t+1}^{T,f}, \tilde{L}_{t+1}^T]/E_t[M_{t+1}]$ and $\psi_t^{T,f} = \text{cov}_t[\Lambda_{t+1}^{T,f}, \tilde{K}_{t+1}]/E_t[M_{t+1}]$ are the covariances of the convenience yield with default risk and regulatory risk, respectively. $\square$

Proof of Proposition 3. Following the same reasoning as in the two previous proofs, the price of an EME sovereign bond issued offshore in dollars, $\hat{P}_t^T$, is given by:

$$\hat{P}_t^T = E_t[M_{t+1}\hat{P}_{t+1}^T\hat{\Lambda}_{t+1}^{T,f}] \quad (36)$$

Assume the local government can default on this bond with probability $\hat{\pi}_t^T$, imposing a loss of $\hat{L}_{t+1}^T$ on the investor. In this case, $\tilde{L}_{t+1}^T$ is a random variable taking the value $\hat{L}_{t+1}^T$ in the case of default and zero otherwise. However, since the bond is issued in dollars and offshore, the government cannot impose capital controls or currency convertibility restrictions. Therefore, assuming again one-period bonds and continuous compounding, the price is given by

$$\begin{aligned} e^{-\hat{y}_t^T} = \hat{P}_t^T &= E_t[M_{t+1}\hat{\Lambda}_{t+1}^{T,f}] - E_t[M_{t+1}]E_t[\tilde{L}_{t+1}^T] - \text{cov}_t[M_{t+1}, \tilde{L}_{t+1}^T] - \text{cov}_t[\hat{\Lambda}_{t+1}^{T,f}, \tilde{L}_{t+1}^T] \\ &\approx e^{\hat{\lambda}_{t+1}^{T,f} - \hat{\pi}_t^T(E_t[\hat{L}_{t+1}^T] + \text{cov}_t[M_{t+1}, \tilde{L}_{t+1}^T]/E_t[M_{t+1}]) - \text{cov}_t[\hat{\lambda}_{t+1}^{T,f}, \tilde{L}_{t+1}^T]/E_t[M_{t+1}]} E_t[M_{t+1}] \end{aligned} \quad (37)$$

Taking logs on both sides gives:

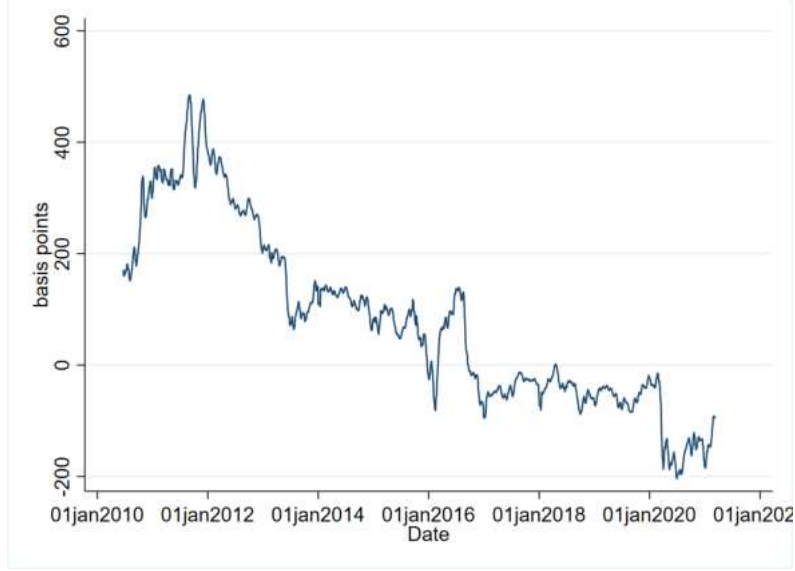
$$\hat{y}_t^T \approx y_{rf,t}^{US} - \hat{\lambda}_t^{T,f} + \hat{l}_t^T \quad (38)$$

where variables have the same interpretation as in the previous two proofs. Now, define Φ_t^{FC} as the spread between the yield of the synthetic bond (Equation (35)) and the yield on the foreign currency-denominated bond (Equation (38)). Then,

$$\begin{aligned} \Phi_t^{FC} &\equiv y_t^T - \rho_t - \hat{y}_t^T \\ &\approx (y_{rf,t}^{US} - \lambda_t^{T,f} + (l_t^T - q_t^T) + (k_t - p_t) - \xi_t^{T,f} - \psi_t^{T,f}) - (y_{rf,t}^{US} - \hat{\lambda}_t^{T,f} + \hat{l}_t^T) \\ &= (\hat{\lambda}_t^{T,f} - \lambda_t^{T,f}) + (l_t^T - \hat{l}_t^T - q_t^T) + (k_t - p_t) - \xi_t^{T,f} - \psi_t^{T,f} \end{aligned} \quad (39)$$

□

Figure 8: Local vs. Foreign jurisdiction spread for Brazil



Notes: The Figure shows the spread between the swapped local-currency sovereign bond and the foreign-currency-denominated bond.

IB. Robustness for Section 2

We describe some financial frictions and prominent features in the markets for EMEs' local-currency government bonds. We address how these issues may affect our estimate of the local-currency convenience yield and propose some robustness checks when applicable.

IB.1. The role of regulatory risk

As explained in the main text, local-currency bonds in EMEs carry the risk of the local government imposing capital controls, taxes on outflows, or currency convertibility restrictions. The term $\Phi_t^{FC,i}$ in Equation (9) intends to account for the risk by taking the spread between sovereign bonds issued under international vs. domestic law. The former does not give as much regulatory freedom to the EME government; therefore, this spread should account for most of these regulatory risks.

In this subsection, we want to provide an idea of how relevant this adjustment is by using the example of Brazil. Figure 8 plots the time series of $\Phi_t^{FC,i}$ for the period 2010-2021.

Recall from Equation (8) that this spread will be larger: (1) the larger the domestic regulatory risk (k_t^j), and (2) the lower the covariance between default and regulatory risk and

currency risk (q_t^j and p_t^j). The spread is positive and large at the beginning of the sample. Importantly, this period coincides with the Brazilian government’s imposition of capital outflow taxes. In October 2009, the government introduced a tax on financial transactions (the IOF) of 2% on foreign investment in fixed-income instruments. In 2010, the tax was raised to 4 and then to 6%, and stayed at that level until it was abandoned in June 2013. Consistent with this timing, the spread $\Phi_t^{FC,i}$ moved around 200-500 basis points. A negative value of this spread (relevant after 2016) means that the positive covariance of currency risk with other risks is more significant than the risk of capital controls and other regulations.

IB.2. Eurobonds

Eurobonds are securities denominated in a different currency than the local one of the country where the bond is being issued (despite their name, they are not necessarily bonds issued in Europe or in euros). EME sovereigns frequently issue Eurobonds, which usually correspond to sovereign bonds issued in international markets in the EME’s local currency.

Importantly, these bonds are governed under international law, settled in U.S. dollars, and therefore free of capital control, convertibility restrictions, and other regulatory risks imposed by the EME government. Equation (9) in Section 2.1 measures the convenience yield for local-currency bonds issued under *domestic* law, and that is the reason it corrects for the risk of capital controls and other regulatory risks imposed by the local government. However, this correction is unsuitable for Eurobonds, and Equation (9) overstates the magnitude of the local-currency convenience yield if a country issues most of its local-currency debt via Eurobonds.

Although we don’t have a precise breakdown of Eurobonds on the total local currency sovereign debt outstanding, we use the International Debt Securities (IDS) database from the Bank of International Settlements to get an estimate of the prevalence of Eurobonds in local-currency sovereign debt in EMEs. The IDS reports the outstanding government bonds issued in international markets in local currency. Although it doesn’t distinguish between foreign and domestic law, it still serves as a proxy for the amount of local currency bonds governed by foreign law.

Table 7 shows the percentage of outstanding local currency government bonds issued in

international markets according to IDS over the total amount of outstanding local currency bonds issued in all markets. Data is available for only 5 of the nine countries in our sample.

Table 7: Share of total LC-bonds outstanding issued in international markets

Country	Mean	Max
Brazil	0.5%	0.9% (Dec. 2007)
Chile	2.4%	4.8% (Dec. 2021)
Colombia	3.6%	6.1% (Dec. 2007)
Peru	35.4%	47.5% (Dec. 2019)
Philippines	3.3%	4.3% (Dec. 2021)

Notes: annual frequency for 2004-2021. Share calculated with outstanding values at the end of each year. Column 3 shows the year in which the maximum share was achieved.

Overall, only Peru has a significant amount of outstanding local-currency bonds issued in international markets as a proportion of total local-currency debt. Brazil has less than 1% of the total, while Chile, Colombia, and the Philippines move around only 3% of the whole. Even if all these local-currency bonds are governed by foreign law, that still would represent a minimal percentage with the only exception of Peru. However, Peru has no capital controls on foreign investments during the period considered.

Regarding the discussion on market segmentation, a second dimension of market segmentation can arise between local-currency bonds issued under international and domestic law. In this case, it might be that the whole share of local-currency government bonds owned by foreigners correspond to bonds governed by international law (Eurobonds), while local investors own only the bonds issued under domestic law. Again, evidence does not show this to be the case. Onen et al. (2023) show that most of the increase in foreign ownership of local-currency sovereign bonds in the past two decades has come from foreigners increasingly participating in the *domestic* market.

IB.3. Analysis of the U.S. Treasury Premium

In this Section, we replicate the analysis of the dollar convenience yield, but with the U.S. Treasuries as the benchmark asset instead of the non-Treasury safe assets. This exercise

resembles the one for G10 countries shown in Du, Im, and Schreger (2018).

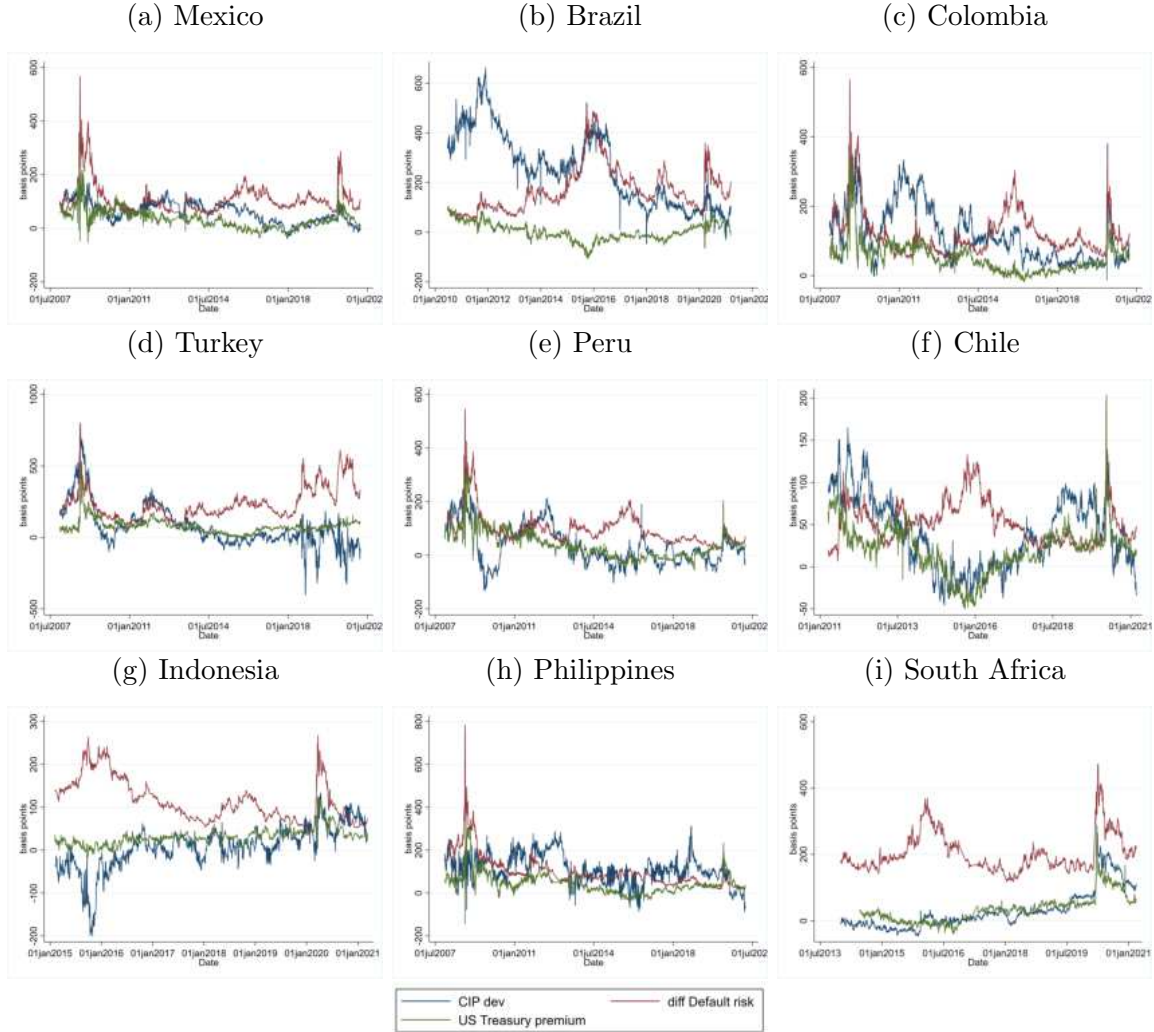
In this case, $y_t^T - \rho_t - y_t^{US}$ corresponds to the CIP deviation between the two sovereign bonds (notice that now we are subtracting the U.S. yield from the swapped EME bond). The term $\lambda_t^{US,f} - \lambda_t^{T,f}$ corresponds to the U.S. Treasury premium (how much investors pay for the safety/liquidity of U.S. Treasuries against EME local-currency bonds). $\lambda_t^{US,f}$ is proxied by the spread between the U.S. agency bond and the U.S. Treasury, and $\lambda_t^{T,f}$ by the dollar convenience yield estimated in Section 2.1.2.

Figure 9 compares the evolution of CIP deviations and two components: differential default risk and the U.S. Treasury premium. CIP deviations spiked during crises (i.e., in 2008 and 2020), which was driven by an increase in differential default risk and the U.S. Treasury premium. The increase in the U.S. Treasury premium aligns with intuition: During financial distress, investors prefer the liquidity and safety of U.S. Treasuries. After 2008, the U.S. Treasury premium steadily declined until 2015-2016. This means that during this period, investors were willing to pay a lower premium for the safety and liquidity of U.S. government debt versus comparable debt of EMEs. This premium then increased again until the end of the sample.

These patterns starkly contrast with the G10 counterparts Du, Im, and Schreger (2018) estimated. In that paper, the authors showed that the U.S. Treasury premium for long maturities became consistently negative after 2010, meaning that investors were no longer willing to pay an extra price for the safety and liquidity of U.S. Treasuries compared to the sovereign bonds of the G10 countries. Based on this result, some authors have cast doubt on the safety status of long-term U.S. Treasuries. Figure 9 shows that this is not the case for EMEs. U.S. Treasuries are still considered a safe asset compared to their EME counterparts.

Surprisingly, CIP deviations outside of financial crises closely followed the U.S. Treasury premium dynamics -and not the dynamics of default risk- for Mexico, Colombia, Peru, Chile, Indonesia, and South Africa. In these countries, even though differential credit risk significantly increased in 2015-16, CIP deviations decreased, following the dynamics of the U.S. Treasury premium. This is surprising as research on EMEs has predominantly focused on the determinants of default risk, not convenience yields. One final note of caution is needed for Turkey in 2018-2019. The series for CIP deviations became very noisy and

Figure 9: CIP Deviation and Components, 5-Year Local-Currency Sovereign Bonds



turned negative. These were years of severe capital outflows and recession in Turkey, and the negative values of the CIP deviation likely arose because of market segmentation, in which only local investors predominantly hold local-currency sovereign bonds.

The role of capital control risk (absent in Du, Im, and Schreger, 2018) can be seen in Figure 9 by the vertical distance between the CIP deviation (blue line) and the two components shown (red and green lines). This was accounted for by the sum of the capital control risk term plus the covariances term Equation (6). Two episodes in the data stand out: Brazil during 2010-2014 and Colombia soon after 2010. In the case of Brazil, the government imposed a tax on financial transactions in October 2009 to curb portfolio investment flows and cross-border derivative trading. Still, the tax was lifted in June 2013.

Table 8: Determinants of Convenience Yields

	(1)	(2)	(3)
	Domestic Conv. Yield	Dollar Conv. Yield	
Dep. var.:	Safety Premium	Liquidity Premium	y_t^{US} : AAA US corp. bonds
$\log(\frac{\text{Local gov debt}}{GDP_{local}})_{t-1}$	-23.31*** (3.895)	-8.376** (3.236)	
$\log(\frac{\text{U.S. gov debt}}{GDP_{US}})_{t-1}$	50.93 (40.42)	-19.54 (30.95)	-290.1** (118.8)
slope $_{t-1}$	-4.125*** (1.517)	4.463*** (1.311)	
1-year CDS $_{t-1}$	0.094*** (0.030)		
Local MP rate $_{t-1}$		3.629*** (1.185)	
U.S. MP rate $_{t-1}$		6.048 (3.856)	-21.54** (8.915)
slope $_{US,t-1}$			-26.85*** (9.745)
Constant	79.75 (62.51)	-106.3 (62.78)	-352.0* (190.4)
Observations	995	1,627	1,137
R-squared	0.701	0.302	0.567

Notes: Data are at monthly frequency. All columns include country and year-fixed effects. Standard errors are double-clustered by country and year. Start dates vary among countries but end in March 2021 for all. U.S. debt and EME local-currency debt-to-GDP variables are net of the central bank's holdings. *** p<0.01, ** p<0.05, * p<0.1

IC. Robustness for Section 3

IC.1. The Role of Safety/Liquidity Services

Table 8 shows the estimation of regression (10) of Section 3.2 with alternative dependent variables. Column 1 replicates regression (10) with the safety premium component as the dependent variable. Here, we omit the monetary policy rate from the independent variables because there is no formal link with the safety premium in the literature (unlike with the liquidity premium). We add the 1-year CDS spread to further control for default risk. Column 2 regresses the liquidity premium component. Overall, both columns show similar results to those in Section 3.2.

Column 3 replicates regression (10) with the ICE index of AAA-rated US corporate bonds as the proxy for y_t^{US} instead of the Refcorp bonds. Results are generally robust to the use

of this alternative dollar safe asset. The coefficient on the supply of US government debt is negative and significant, as in Section 3.2. The issue with this ICE index is that it combines corporate debt of all maturities greater than one year. In this case, it also captures variation in the US term premium. The negative sign of the monetary policy rate can relate to this issue.

IC.2. Credit risk as the dependent variable

If the estimation of convenience yields in Section 2.1 successfully disentangled differential default risk from differential convenience yields, then default risk should respond differently to the determinants of convenience yields analyzed in Section 2.3. The representative-agent asset pricing model says that standard credit risk does not depend on the supply of debt.

Table 9 replicates the regressions in Section 2.3 but with the CDS for each country as the dependent variable. The larger number of observations is because we had data for CDS spreads for a few more countries than we had convenience yield estimates. Unlike the EME local-currency convenience yield, credit risk is unaffected by the supply of government debt, suggesting that the convenience yield accurately captures the demand for safety and liquidity. The local monetary policy rate level increased credit risk since it likely increased the cost of servicing the debt. The VIX index also positively impacted credit risk, which is consistent with intuition. Interestingly, debt inflows to government debt significantly reduced credit risk, which is expected as foreigners' buying local debt increases the chance of repayment. The same happened with inflows into bank debt, which is consistent with sovereign debt being mostly held by banks in EMEs.

IC.3. Domestic Conv. Yields and Risk Factors (3-month maturity)

To further address the concern about the presence of standard risk premia, we replicate Table 3 in the main text for the 3-month maturity, where arguably risk premia are very small. Table 10 shows the results.

Results hold, although with lower magnitudes. Notably, although the safety premium still responds positively to global risk, it now correlates negatively with local risk, suggesting

Table 9: Determinants of Credit Risk (5-Year Sovereign Bond)

Dep. var: $cds_{i,t}$	(1)	(2)	(3)	(4)	(5)
MP rate $_{t-1}$	11.94*** (1.410)	11.48*** (1.377)	11.65*** (1.404)	11.70*** (1.413)	6.214*** (1.924)
$\log(\frac{\text{US debt to GDP}}{\text{Debt to GDP}})_{t-1}$	-8.027 (16.09)	-4.336 (14.86)	-8.776 (16.37)	-9.849 (16.47)	-30.78** (12.11)
US fed funds $_{t-1}$	-14.66** (7.115)	-11.31 (7.349)	-13.86* (7.149)	-14.09* (7.177)	-11.77 (7.894)
vix $_{t-1}$	4.575*** (0.420)	4.352*** (0.456)	4.429*** (0.421)	4.339*** (0.442)	4.271*** (0.520)
K flows	N	Y	Y	Y	Y
ToT	N	N	N	Y	Y
Inflation	N	N	N	N	Y
Constant	-1.286 (88.08)	-28.20 (86.53)	-7.972 (91.53)	1,110 (875.5)	1,058 (746.5)
Observations	1,338	1,338	1,338	1,338	1,213
R-squared	0.689	0.702	0.698	0.700	0.734

Notes: All columns include country and year-fixed effects. *** p<0.01, ** p<0.05,

* p<0.1

that short-term uncertainty may be more relevant than medium-term uncertainty.

On the other hand, the liquidity premium at the 3-month maturity responds positively to global risk, which suggests that very short-term government debt shares more money-like properties than medium-term bonds. This is the case of the US, where it is well known that short-term T-bills are the most money-like in the term structure (Greenwood, Hanson, and Stein, 2015; Du, Im, and Schreger, 2018).

Table 10: Domestic Convenience Yield and Risk Factors (3m Maturity)

Dep. var.: dom. CY	(1)	(2)	(3)	(4)
A. Safety Premium				
VIX _{t-1}	-0.065 (0.078)		0.170** (0.067)	
mar _{t-1}		-0.901 (1.050)		2.013** (0.894)
Economic risk _{t-1}	-3.626*** (0.810)	-3.624*** (0.809)		
Financial risk _{t-1}			-4.325*** (0.817)	-4.321*** (0.801)
Observations	1,063	1,063	1,063	1,063
R-squared	0.792	0.792	0.794	0.794
B. Liquidity Premium				
VIX _{t-1}	-0.357 (0.293)		1.035*** (0.396)	
mar _{t-1}		-5.440 (3.492)		10.20* 5.180
EPU _{t-1}	11.12*** (2.892)	11.12*** (2.873)		
Financial risk _{t-1}			-1.704 (2.150)	-1.338 (2.253)
Observations	595	595	1,229	1,229
R-squared	0.380	0.380	0.355	0.352
Country & Year FE	Y	Y	Y	Y
MP rate & Debt supply	Y	Y	Y	Y
K inflows	Y	Y	Y	Y

Notes: *mar* is the global risk factor in Miranda-Agrippino and Rey (2020). *EPU* is the economic policy uncertainty index of Baker, Bloom, and Davis (2016) -available for fewer countries. Regressions exclude data for Turkey and the Philippines. Data are at a monthly frequency. Standard errors are double-clustered by country and month. Start dates vary among countries, but ICRG data is available until April 2019. *** p<0.01, ** p<0.05, * p<0.1